SIEMENS Healthineers

MAGNETOM

Maintenance Instructions System

Preventive Maintenance

Avanto
Espree
The protocol M6-020.832.05.03.XX is required for these instructions

Document Version

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1.1 General information

To ensure the continuous safety, quality, and reliability of these products, periodic maintenance must be performed in accordance with the Maintenance Instructions / Safety-related tests.

Periodic maintenance includes:

- Safety-related tests (not included in these Maintenance Instructions)
- Preventive maintenance

1.2 Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

1.3 Documentation

A protocol of each maintenance activity must be generated. All fields must be completed.

Each work step or test performed successfully has to be marked "OK" in the test protocol. The designation n.a. (not applicable) indicates that the checkpoint or measured value is not used for this system or that the checkpoint is not due (e.g. interval is more than one year). If an error occurs that cannot be eliminated, mark the respective test with "not OK".

The Maintenance Protocol has to be archived by the operator in the System Owner Manual. We recommend that country organizations also compile such documentation to ensure optimum product operation, e.g. should damage occur.

1.4 Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

1.5 Time Frame for the Maintenance Schedule

Tab. 1 Maintenance Schedules

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Cooling	12	10	
ACC Cabinet	12	10	5-mm Allen key, flashlight
Gradient filters	12	10	
Vacuum pump	24	5	Filter material number 10496420
SEP (optional)	12	20	10 mm open-end wrench, 38/47mm strainer spanner, bucket, water hose for filling
Cold head	12	5	
Adsorbe replacement Compressor CSW 71 Compressor F70	 24 36	70	 Adsorber p.n. 8396009 Adsorber p.n. 7461127
Computer	12	10	Vaccum cleaner
TFT monitor	12	10	
Patient table	12	15	4mm Allen key
Patient Trolley	12	10	2 batteries (type "AA")
Software	12	30 - 60 [1]	CD-R with medical grade
Annual average time including all options [2]: 158 min (2.63 h)			

1. Depending on data quantity (calculation with 30 min)
2. Without cold head exchange, without mobile systems

Quality Assurance (QA) is performed during safety-related tests (SRT).

The RF cabin and the door of the RF cabin are not part of the MAGNETOM system maintenance and not considered in these instructions. The RF cabin and the RF cabin door have to be maintained according to the instructions of the manufacturer.

Tab. 2 Maintenance Schedules Mobile Systems

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Mobile (Trailer) Systems	12	60	Flat bladed screwdriver (Non-magnetic)
Annual average time including all options : 60 min (1 h)			

Tab. 3 Cold Head

Component / activity	Interval (month)	Time required (min)	Note
Cold head replacement	see note[1]	240	Special training is required for the cold head replacement. The cold head replacement procedure is not included in this description [2].

1. The cold head has to be replaced latest every 24 months unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible.
2. Service personnel training on the Sumitomo cold head is offered by SMT. Instructions for cold head replacement are shown in the video "Sumitomo Refrigerator System". The video is a supplement to the SMT course.

2.1 System Status

- Switch System off

In case to access to the components installed on the magnet the appropriate service covers have to be removed, please refer to REP - System/Covers.

2.2 Cooling

Required time: 10 min

Required tools: n.a.

Prerequisite: System is switched on

PM Cooling system general checks

Check the entire water system for leaks and condensation, especially all water connections and water hoses within the:

- ACC
- GPA
- ACS
- Cold head compressor
- SEP or IFP (option)
- RF-filter plate
- TAS
- Magnet

Service Software

- Open > Service Software > Magnet & Cooling > Cooling Status
- Check the cooling status of the system
  - Cabinet
  - ACS
  - SEP or Chiller / IFP

2.3 ACC Cabinet

Required time: 10 min

Required tools: 5-mm Allen key, flashlight

PMP Checking the fan assembly

The fan assembly is integrated into the ACC cabinet.

- Switch off the GPA and ACC cabinets.
- Carefully swivel out the electronic section of the ACC.

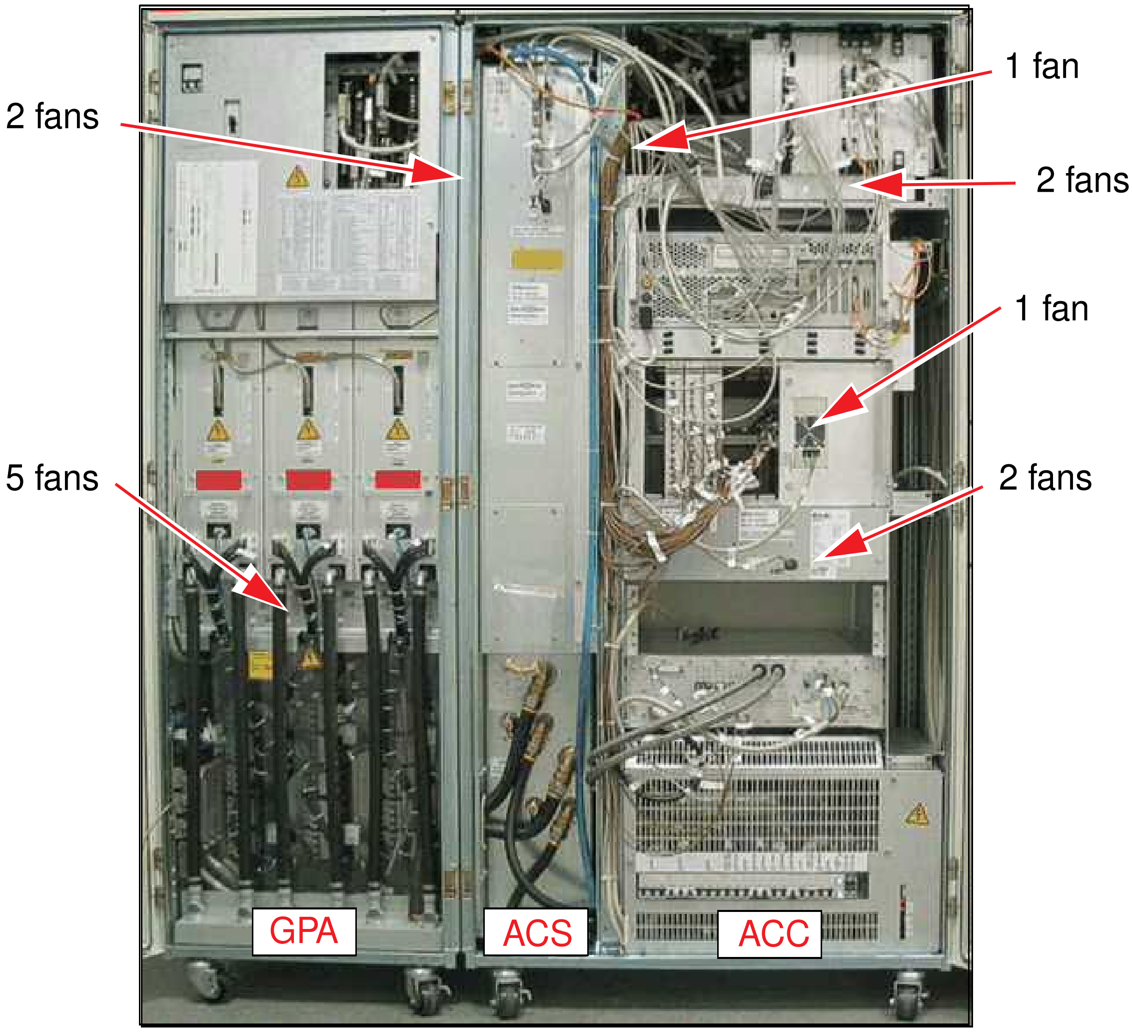
WARNING

High-voltage!

Contact with live parts may result in death or serious injury.

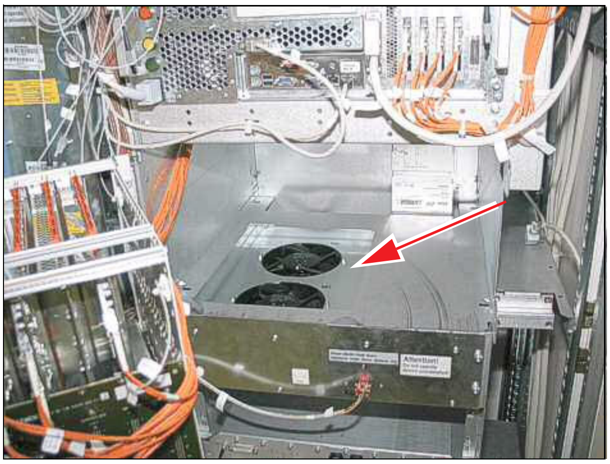
» **Keep away from live parts**

Fig. 1: fans GPA, ACS, ACC



- Switch on the GPA and ACC cabinets.
- Visually inspect the operation of the fans in the ACC cabinet by using a flashlight. Replace the parts if necessary.

Fig. 2: Fans in ACC view



CAUTION

Fans in operation

Sharp edges may cause injury.

» **Do not touch fans during operation.**

- Switch off the GPA and ACC cabinets.
- Carefully swivel in the electronic section of the ACC.

The fans of the GPA are monitored. There is no need to check.

2.4 Gradient filters

Required time: 10 min

Required tools: n.a.

The gradient filters are ventilated by 3 fans per axis.

PM Checking the gradient filter fans

Fig. 3: Gradient filter fans

» If one or 2 fans per axis are defective, replace the fan assembly.

» If all 3 fans per axis are defective, the corresponding gradient filter has been damaged due to overheating. Replace the fan assembly **and the corresponding gradient filter.**

2.5 Vacuum pump (comfort kit option)

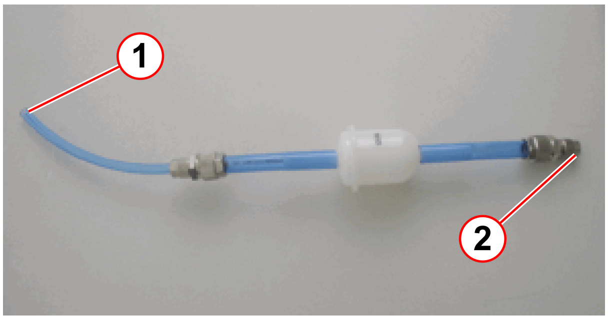
Required time: 5 min

Required tools: Filter

PMP Replacing the vacuum pump filter (every 2 years)

The vacuum pump is an option (comfort kit) mounted on the RF filter plate.

Fig. 4: Vacuum pump filter



(1) Connection - vacuum pump
(2) Connection - hose to the examination room

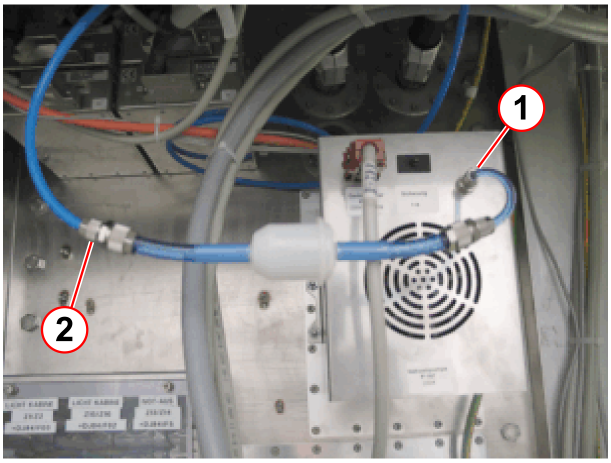
It is no longer necessary to replace the filter inside the pump. An external filter is now available. The external filter prevents the pump and the filter inside the pump housing from dirt.

The old filter can remain inside the pump housing.

- Check the filter inside the vacuum pump if the throughput is insufficient.
  » Replace the filter if required

If no external filter is installed, install the external filter in the same manner as described below.

Fig. 5: External air filter of the vacuum pump



(1) Coupling nut - vacuum pump
(2) Coupling nut - hose to the examiation room

- Loosen the coupling nut (1) at the housing of the filter pump.
- Remove the hose.
- Remove the coupling nut and place it on the hose of the new filter.
- Connect the new filter to the vacuum pump and mount it with the coupling nut.
- Loosen the coupling nut (2) of the hose to the examination room.
- Remove the old filter.
- Connect the new filter to the hose of the examination room and mount it with the coupling nut.

- Enter the replacement date in the Maintenance Protocol.

2.6 SEP (option)

Depending on installed chiller type (customer related) it may be necessary to switch off and refill the chiller. Therefore it is recommended to plan the SEP maintenance together with the chiller maintenance.

Required time: 20 min

Required tools: 10mm open-end wrench, bucket, water hose for filling, 38/47mm strainer spanner p.n. 8116456; if industrial or process water connection is not available a water pump is needed, e.g. manual pump p.n.8115136.

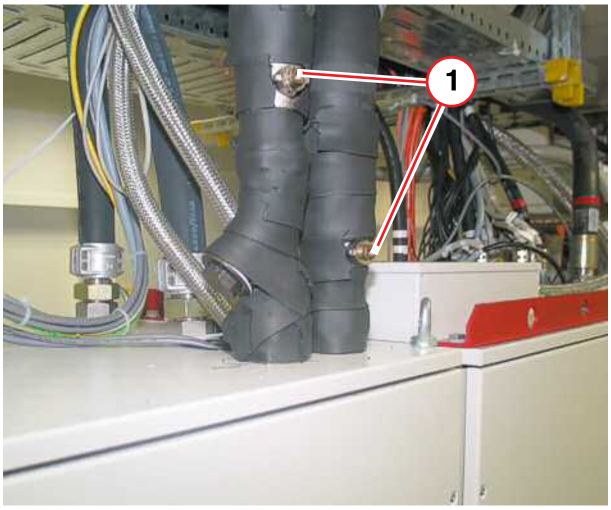
Prerequisite: System is switched off

Abbreviation: clockwise - cw; counterclockwise - ccw

> Perform the maintenance steps
> - Cleaning the strainer of the primary water circuit
> - Cleaning the strainer of the secondary water circuit
> at once.

- Switch off F102 on the top of the ACC cabinet.
- Acknowledge the alarm on the alarm box.

Fig. 6: SEP roof

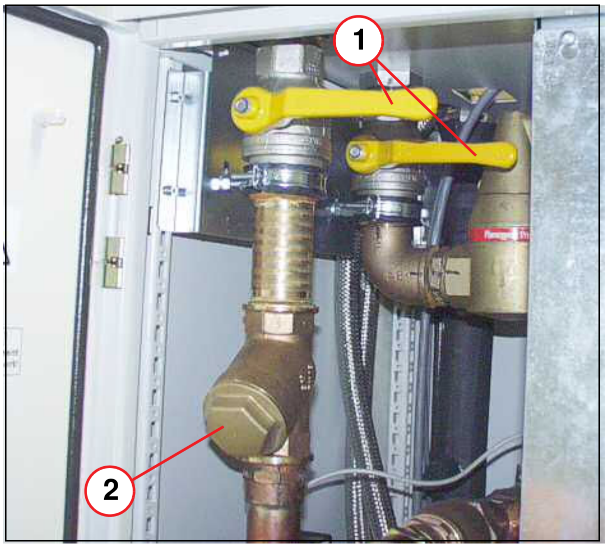


(1) Shut-off valves (primary water circuit)

Close the shut-off valves (1) to isolate the primary water circuit.

- Turn the shut-off valves ccw by 90° using a 10mm open-end wrench.

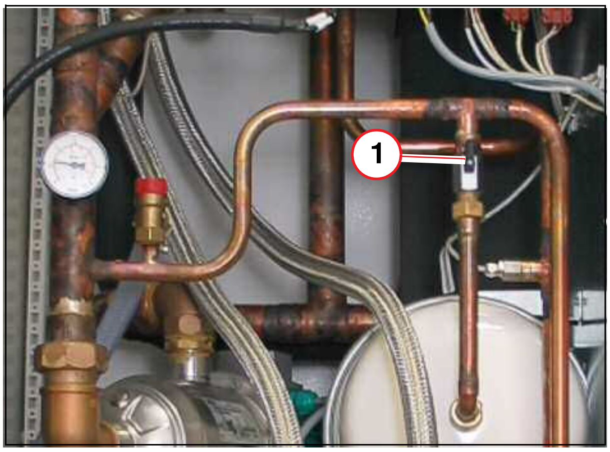
Fig. 7: Shut-off valve/Filter (secondary water circuit)



Close the shut-off valves (1) to isolate the secondary water circuit.

- Turn the shut-off valves ccw 90°.

Fig. 8: Expansion chamber

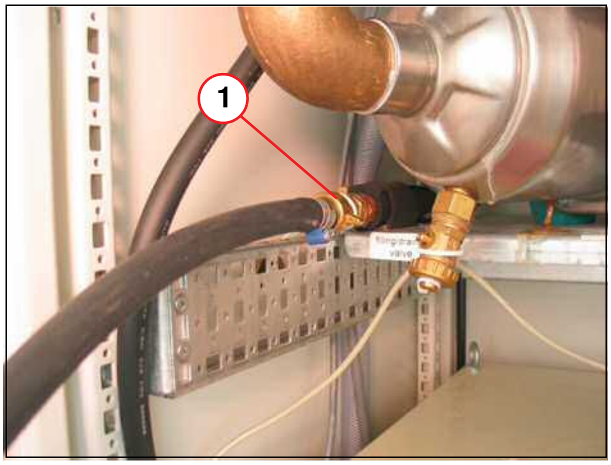


(1) Valve

- Close the valve at the expansion chamber, by rotating knob 90 degrees CW.

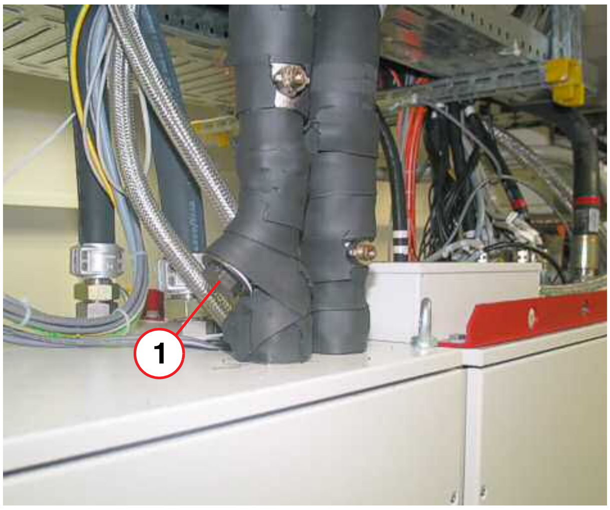
PMP Cleaning the strainer of the primary water circuit

Fig. 9: Primary filling and drain port



- Connect the water hose to the primary filling and drain port (→ Fig. 9 Page 18)and run the hose to a drain container.
- Open the drain port and drain The estimated volume of drained water is approx. 3 liters.
- Carefully loosen the strainer cap using a 41 mm strainer spanner to speed up the draining process ((→ 1/Fig. 10 Page 18)primary water circuit).
- Close the drain port.

Fig. 10: SEP roof



(1) Strainer (primary water circuit)

- Remove the strainer by using a 41 mm strainer spanner ((→ 1/Fig. 10 Page 18)primary water circuit).
- Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the strainer and tighten the cap with the 41 mm strainer spanner.

PMP Cleaning the strainer of the secondary water circuit

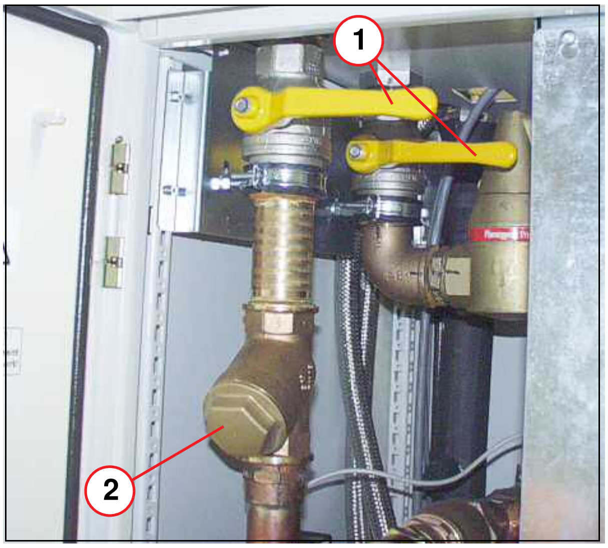
Fig. 11: Secondary filling and drain port

- Connect the water hose to the secondary filling and drain port and run the hose to a drain container.

The estimated volume of drained water is approx. 3 liters.

- Open the drain port and drain.
- Carefully loosen the strainer cap using a 50 mm strainer spanner to speed up the draining process (secondary water circuit).
- Close the drain port.

Fig. 12: Shut-off valve/Filter (secondary water circuit)



- Remove the strainer cap (2) by using a 47 mm strainer spanner (secondary water circuit). Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the strainer and tighten the cap with the 45 mm strainer spanner.

- Open the shut-off valves for the primary (→ 1/Fig. 10 Page 18) and secondary water circuit (→ Fig. 12 Page 19).
- Open the valve of the expansion chamber (→ 1/Fig. 8 Page 17).

Fig. 13: Water filling

- Connect the water hose to industrial or process water, or use a hand pump, or pump with a drill.

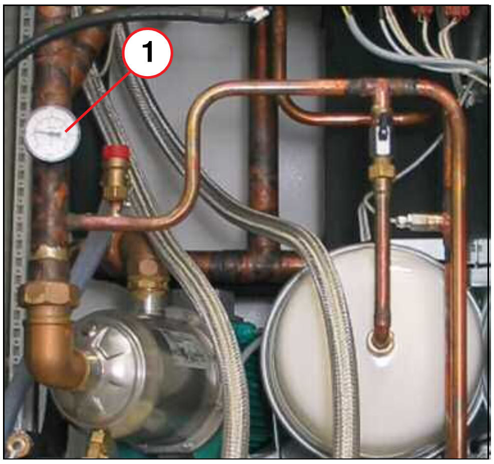
> **i** Do not use any connection to a drinking water system!

Fig. 14: Secondary filling and drain port

- Connect the water hose from the on-site water supply to the filling port of the SEP.

Note: Before connecting the hose to the SEP, the hose should be filled with water and not contain air.

Fig. 15: SEP manometer



- Open the drain valve.
- Refill the SEP with water until the filling pressure is reached (➔ 1/Fig. 15 Page 21).

  » The filling pressure is approximately 0.5 - 1.0 bar (➔ 1/Fig. 15 Page 21).
- Switch on F102 on the top of the ACC cabinet.

> i It takes up to 2 minutes until the water pump is running.

- Close the drain valve and disconnect the water hose. Drain the water from the fill hose into the bucket or sink.

  » The cold head is running.
- Check for leaks at the primary and secondary strainer caps after SEP pump is operating.

2.7 Magnet System

2.7.1 Cold Head

Required time: 5 min

Required tools: n.a.

The cold head has to be replaced latest every 24 months unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible.

PM Cold head is monitored (cold head replacement on demand)

PM Replacing the cold head (w/o cold head monitoring replacement every 24 months)

The cold head has to be replaced **every 24 months** according to OEM recommendation.

> Special training is required for the cold head replacement. The cold head replacement procedure is not included in this description. During the cold head replacement the magnet pressure switch must be checked.

- After the cold head is replaced, enter the replacement date in the protocol.
- Confirm completion of the magnet pressure switch check in the protocol.

2.7.2 Adsorber

2.7.2.1 Replacing the adsorber (Sumitomo CSW 71; every two years)

Required time: 70 min.

Required parts and tools:

PM Replacing the adsorber (Sumitomo CSW 71)

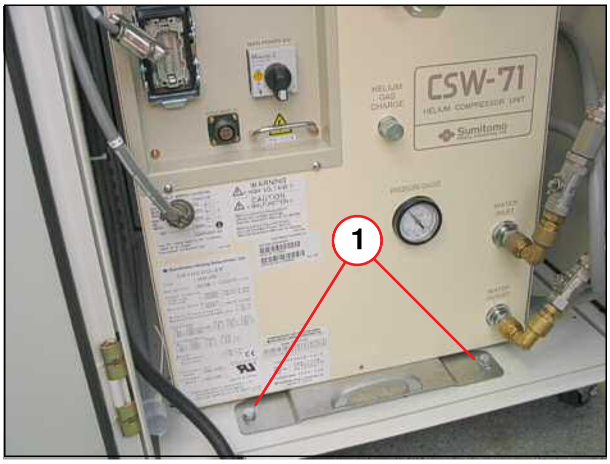
- Adsorber (part number: **8396009;** includes venting adaptor)
- Medium Philips screwdriver, large Philips screwdriver
- Set of nuts
- Fork wrench: 13mm
- Fork wrench: 1”; 1 3/16”; included with magnet accessories

> Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

1. Switch off F102 on the top of the ACC cabinet.
2. Acknowledge the alarm on the alarm box.

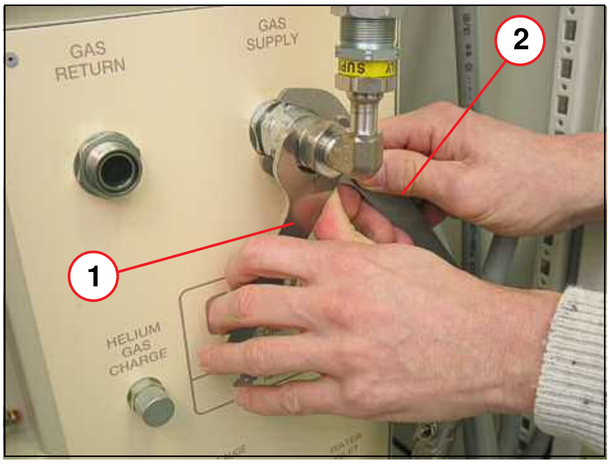
3. Unplug the cables.

Fig. 16: Compressor attachment



- Remove the screws (1).

Fig. 17: Helium flexline coupling



(1) Hold (1” fork wrench)
(2) Loosen (1 3/16” for wrench)

- Remove the self-sealing helium flex-lines.
- Hold nut (1) and loosen nut (2).

NOTICE

Prevent equipment failure.

» **To prevent oil from flowing into unwanted places, do not tip the compressor by more than 10 degrees horizontally.**

Fig. 18: Compressor

The compressor should be pulled onto a sturdy support (e.g., wood) approximately 12 cm in size.

The weight of the compressor is approximately 100 kg.

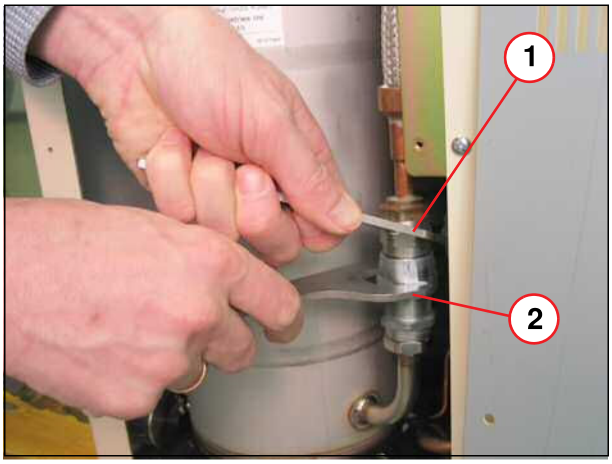
Fig. 19: Compressor out

- Carefully pull out the compressor.
- Remove the cover on the right hand side.

Fig. 20: Compressor, front panel

- Remove nut.

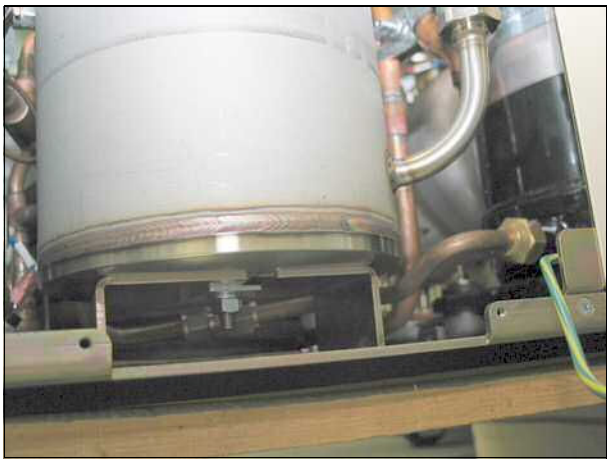
Fig. 21: Adsorber coupling



(1) Hold (1" fork wrench)
(2) Loosen (1 3/16" fork wrench)

- Unscrew the self-sealing coupling.
- Hold nut (1) and loosen nut (2).

Fig. 22: Adsorber mounting screw



- Loosen the 13 mm nut and remove the adsorber.

Fig. 23: Adsorber

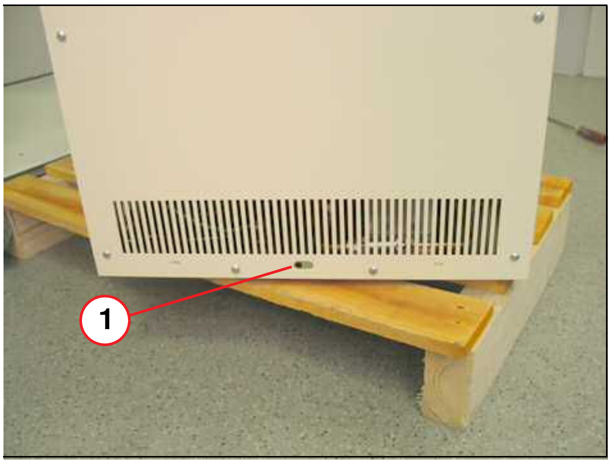
- Install the new adsorber.

> ℹ Tighten all Aeroquip screw connections to the end stop and turn them back by 1/4 turn.

- Insert and tighten the mounting nut to fasten the new adsorber.

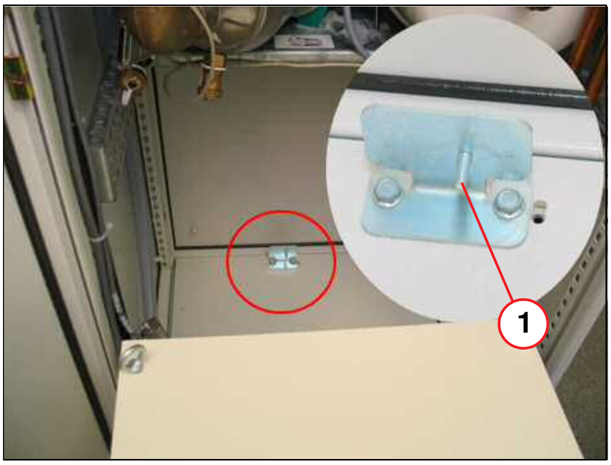
- Connect the adsorber self-sealing coupling (➔ Fig. 21 Page 25).
- Mount the outlet coupling nut (➔ Fig. 20 Page 24).
- Mount the compressor cover.
- Push the compressor into the SEP.

Fig. 24: Compressor, back view



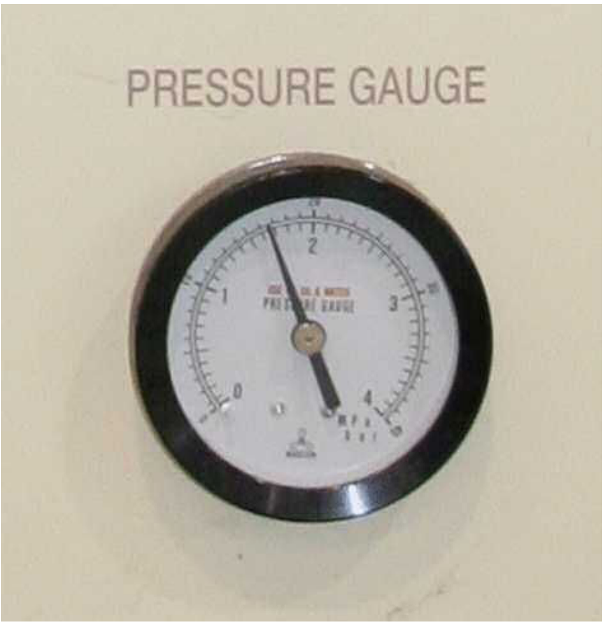
(1) Hole

Fig. 25: SEP - Compressor guide pin



- Slide the compressor guide pin in the compressor hole.
- Secure the compressor in the SEP.
- Connect the cables.
- Switch on F102 on the top of the ACC cabinet.

Fig. 26: Static pressure reading



- Check that the **static** pressure on the compressor gauge is **16.0 - 16.5 bar** (or 1.60 - 1.65 MPa, or 232-239 psi (G).
- Charge with helium gas if pressure is lower.

Tab. 4 Helium gas pressures

Cold head	Static pressure at 20 ^{0}C (680 F)	Dynamic pressure
4 K - 50/60 Hz	16.0 - 16.5 bar 232 - 239 psi (G) 1.60 - 1.65 MPa	21.0 - 23.0 bar 304 - 333 psi (G) 2.10 - 2.30 MPa

- Check the (**dynamic**) operating pressure (refer to the helium compressor pressure gauge).
  » The operating pressure varies according to the heat load of the cold head and temperature around the equipment.

Re-setting messages on the compressor LCD display

- Press the **STOP** button on the LCD display of the compressor.
- **Simultaneously** press and hold the **UP** & **DOWN** arrows.
- While holding the **UP** & **DOWN** arrows, turn **OFF** & **ON** the **Main Power Switch.**
- **Check** the message **does not** appear on the display, and that the compressor **is running**.

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

» **Follow the venting procedure for safe disposal of the used adsorber .**

Safe disposal of the used adsorber

A **venting adapter** is included with the new adsorber.

- **Attach** the **adapter** to one of the **self-sealing couplings** on the used adsorber
- **Vent** the used adsorber to **atmospheric pressure.**
- Turn the **used adsorber** over to a **disposal company.**

2.7.2.2 Replacing the adsorber (F70; every three years)

PM Replacing the F70 adsorber (every three years)

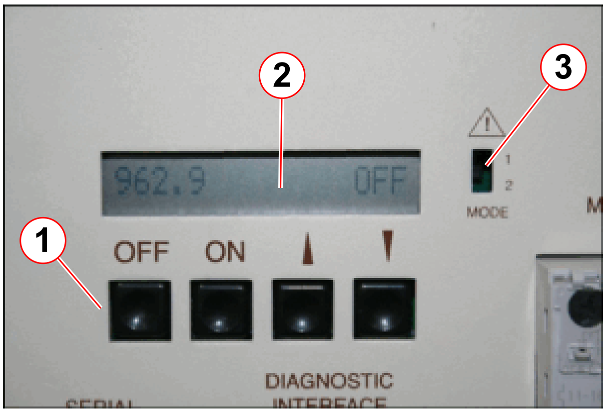
Required time: 70 min

Required parts and tools:

- Adsorber part number: (**07461127;** includes venting adaptor)
- #2 Phillips screwdriver
- F70 Spanner kit **10125650** (delivered with the system)

Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

Fig. 27: F70 System status display



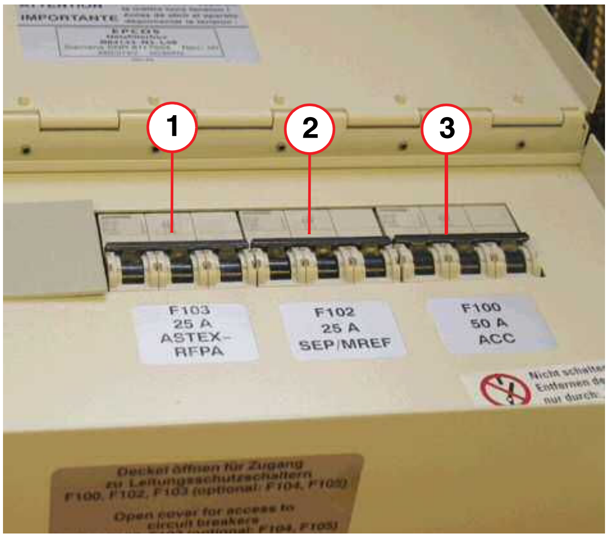
(1) Control buttons
(2) LCD display
(3) Configuration selector switch

- If the **configuration selector switch** has been set to **configuration 2**, both **ON** and **OFF** buttons are **disabled.** (the compressor is controlled via the pressure within the magnet).
  - Switch the compressor **ON** or **OFF** using the **main power switch**.

Adsorber removal

- The compressor may be a **stand-alone** unit, or fitted into an **SEP cabinet**.
- **Press** the **OFF** button on the **control panel**.
- Turn **off** the compressor at the **main power switch.**

Fig. 28: ACC cabinet - Circuit breakers F100, F102, F103

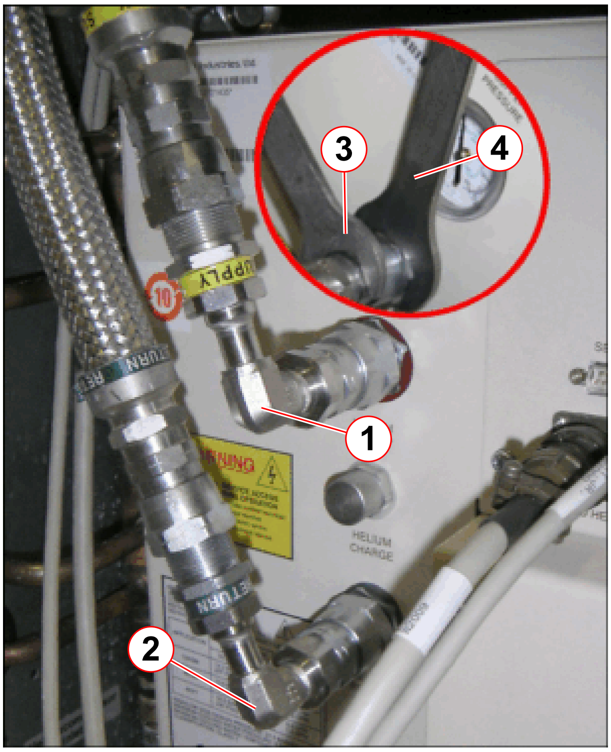


(1) F103 - isolates RFPA
(2) F102 - isolates SEP or IFP (stand-alone compressor)
(3) F100 - isolates ACC cabinet

- **Isolate power** to the compressor:
  » **ACC cabinet** - SEP or IFP (stand-alone unit) - switch off circuit breaker **F102**.

- The **alarm box** will display a red warning LED and an **audible alarm** will sound.
- **Press** the **ACKNOWLEDGE** button to cancel the audible alarm.

Fig. 29: F70 He compressor connections



(1) SUPPLY (high pressure) 90-degree elbow
(2) RETURN (low pressure) 90-degree elbow
(3) Hold nut (1" spanner/wrench)
(4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

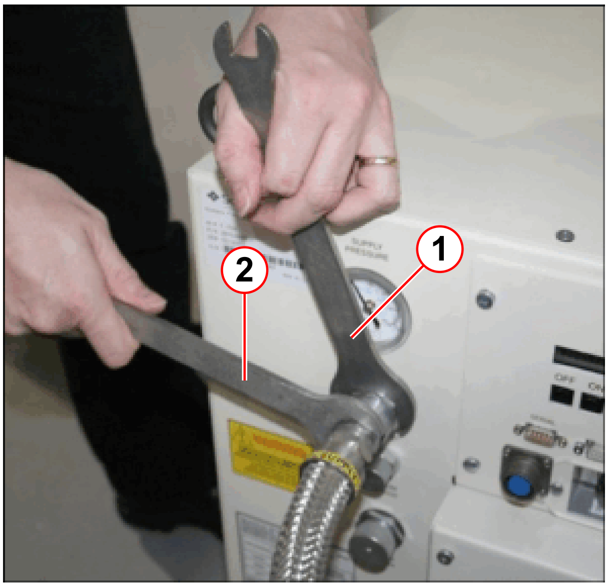
SEP cabinet option

Each **helium pressure line** is connected to the compressor via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
  - **Return** (low pressure)
  - **Supply** (high pressure)
- **Hold** nut (3) and **turn** nut (4).

> 90° elbow connections are not required for a stand-alone unit.

Fig. 30: F70 He compressor connections (stand-alone unit)

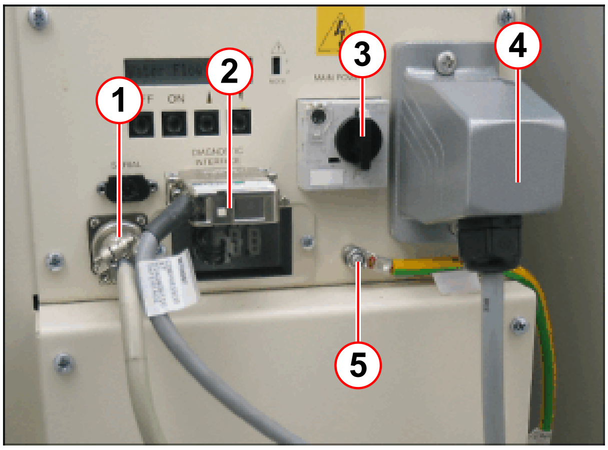


(1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
(2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Remove** the self-sealing helium pressure lines in the following order:
  - **Return** (low pressure).
  - **Supply** (high pressure).
- **Hold** nut (2) and **turn** nut (1).

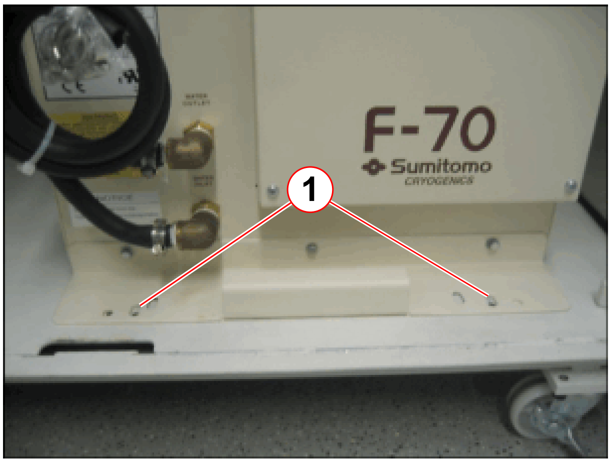
Fig. 31: F70 Electrical cable connections



(1) Cold head cable
(2) MSUP DB25 Diagnostic interface cable
(3) Main power switch
(4) Mains supply cable
(5) Ground wire

- **Disconnect** the **line power supply, cold head motor,** and **MSUP** interface cables.
- **Disconnect** the **ground wire**.

Fig. 32: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

SEP cabinet option:

- **Remove** the 2 M8 hex head screws securing the **fixing plate** to the cabinet.
- Use the **fixing plate handle** to carefully pull the compressor onto the wooden support 'ramp'.

  The unit is fitted with **nylon slide rails** to simplify movement.

WARNING

Risk of personal injury and damage to equipment.

The weight of the compressor is 100 kg (220 lbs).

» **When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.**

NOTICE

Prevent equipment failure.

» **To prevent oil from flowing into unwanted places, do not tip the compressor more than 10 degrees from the horizontal.**

Fig. 33: Supporting ramp

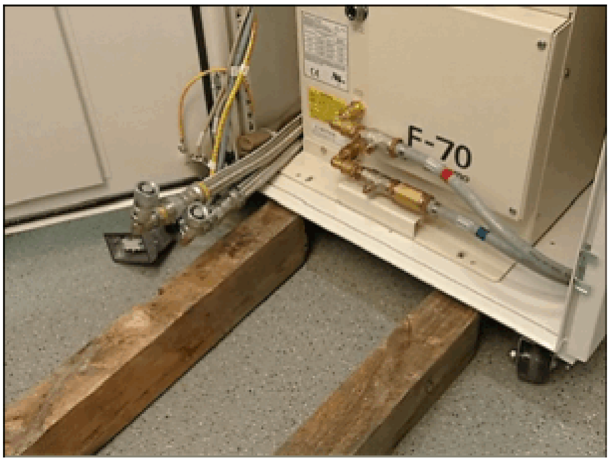
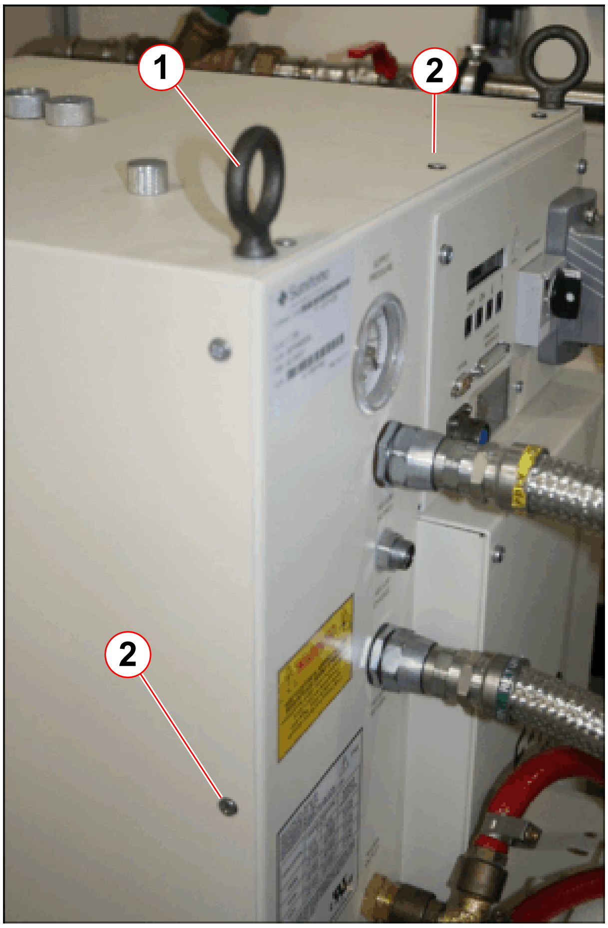


Fig. 34: Move the compressor onto the supporting ramp

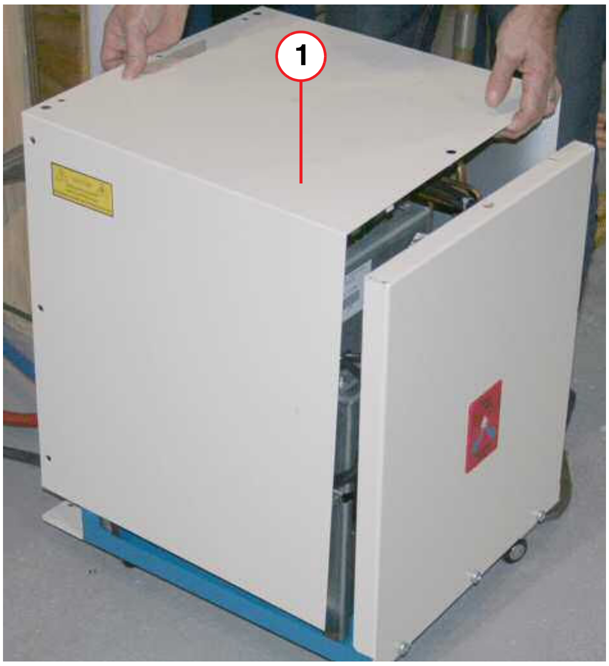
Fig. 35: F70 Lifting eyebolts and cover screws



(1) Lifting eyebolts (3 positions)
(2) Cover panel securing screws

- **Remove** the **3** lifting **eyebolts** and all cover panel **screws**.

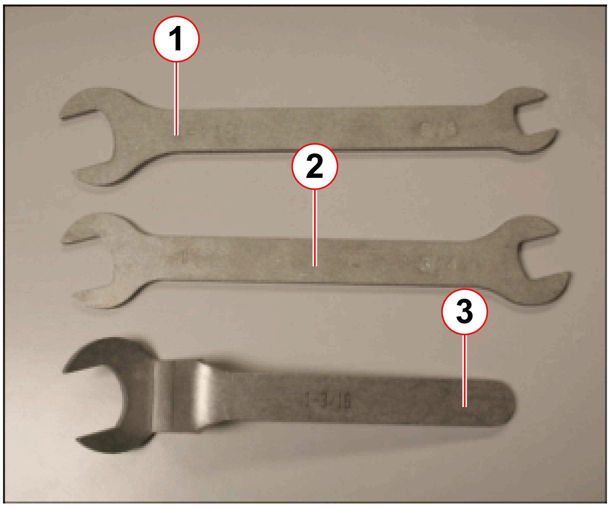
Fig. 36: F70 Removing/replacing the cover



(1) U-shaped cover

- **Lift off** the U-shaped **cover panel.**

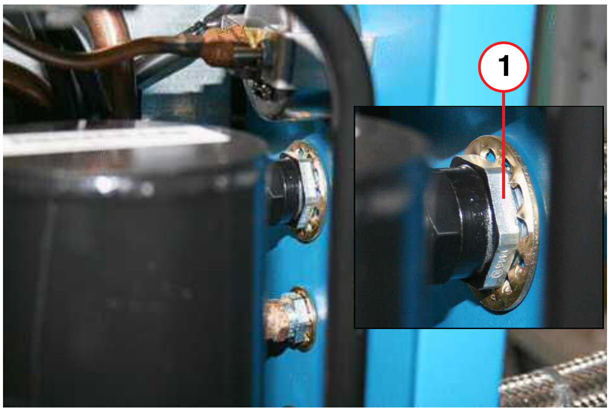
Fig. 37: F70 Spanner kit (part no. 10125650)



(1) 1 1/8" open-end and 5/8" open-end spanner/wrench
(2) 1" open-end and 3/4" open-end spanner/wrench
(3) 1 3/16" open-end crowfoot spanner/wrench

- Always use the three open-ended wrenches when connecting and disconnecting the couplings.
  - **Hold** the stationary nut(s).
  - **Turn** the moveable coupling.

Fig. 38: F70 Adsorber - high pressure supply coupling (inlet side)



(1) HOLD stationary nut (1” open-end spanner/wrench)

To remove the adsorber from the compressor cabinet

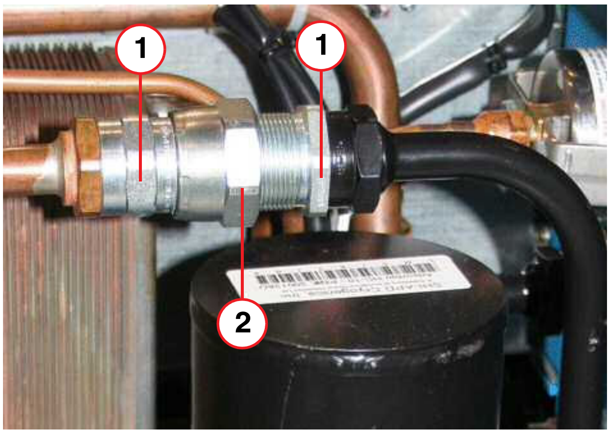
- **Hold** the stationary nut (1) on the **inside**.

Fig. 39: F70 High pressure Aeroquip coupling

(1) Locknut
(2) Flat rubber gasket
(3) Nylon washer

- **Remove** the high pressure **locknut** and **nylon washer** from the front panel.

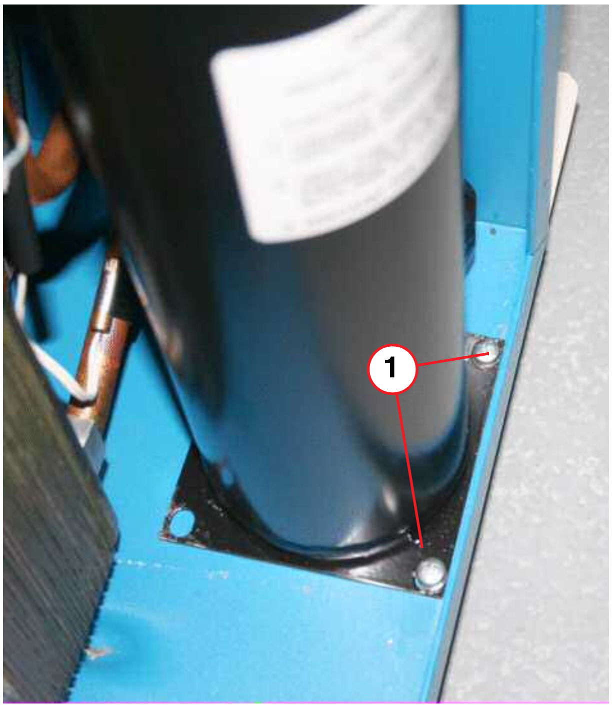
Fig. 40: F70 Adsorber - oil separator Aeroquip coupling



(1) HOLD stationary nut (1” open-endspanner/ wrench)
(2) TURN moveable coupling (1 3/16” open-end crow foot)

- **Disconnect** the adsorber at the **oil separator coupling**

Fig. 41: F70 Adsorber base securing screws



(1) Securing screws (2 off)

- **Remove** the 2 **base securing screws**.
- Carefully **remove** the adsorber.
- **Retain** all removed **attachments**.

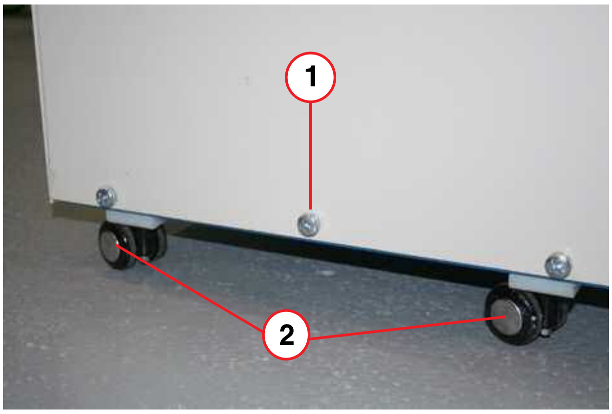
> Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

Adsorber replacement

- **Remove** the **dust caps** from the **aeroquip connections** of the new adsorber.

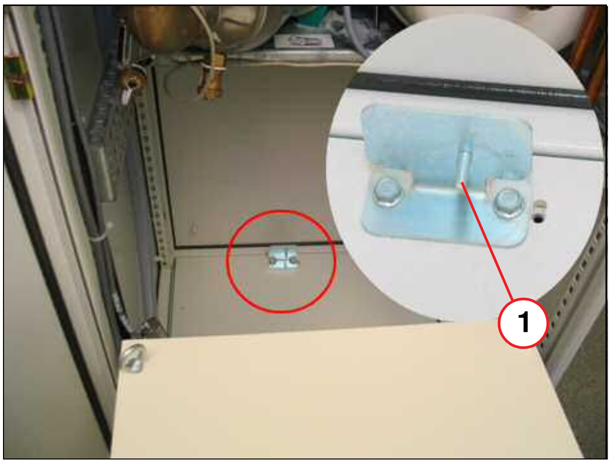
- Lift the adsorber into position inside the compressor.
- **Fit** the **lock washer** on the **aeroquip supply coupling** of the new adsorber.
- **Insert** the **supply coupling** through the **front panel** and position the adsorber.
- Locate the two base **securing screws** to position the adsorber. **Do not fully tighten** at this stage. (→ 1/Fig. 41 Page 35)
- **Fit** the **nylon washer** and **locknut** on the **high pressure supply** coupling at the front panel (→ Fig. 39 Page 34).
- **Connect** the adsorber at the **oil separator coupling** and perform a **leak check.** (→ Fig. 40 Page 35).
- **Fully tighten** the 2 off base securing screws
- **Reinstall** the **cover panel** using the original retaining **screws**.

Fig. 42: F70 Back panel view



(1) Center screw is removed for SEP cabinet installation
(2) Casters x 4 (stand-alone units only)

Fig. 43: SEP - Compressor guide pin

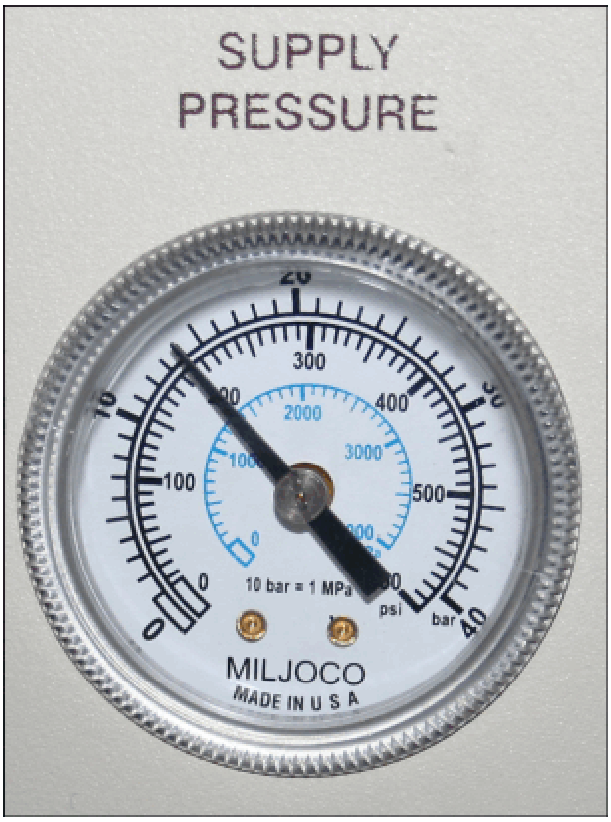


- Slide the compressor into the **SEP cabinet** and locate the **guide pin** in the **compressor hole.**
- **Resecure** the **fixing plate** using the 2 M8 hex head screws.

> Tighten all Aeroquip screw connections to the end stop and turn them back by 1/4 turn.

- **Reconnect** the high pressure (**supply**) and low pressure (**return**) helium gas lines (→ Fig. 29 Page 29).
- Reconnect the **cold head cable, diagnostic cable, and power cable** and **ground wire** (→ Fig. 31 Page 30).

Fig. 44: F70 Static pressure reading



- **Check** the **equalization (static)** pressure on the pressure gauge.
  » **Gas charging** will be required if pressure is below expected.

Equalization (static) pressure at 20 ^{0}C (680 F) for gas lines 12 to 20 meters long

Tab. 5 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- Switch **on F102** (ACC cabinet) (→ 2/Fig. 28 Page 28).
- Switch **on** the compressor at the **main power switch**.
- Check the **alarm box status** and **clear** any **faults**.
- Check the gas Aeroquip **couplings** for **leaks**.

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

» **Follow the venting procedure for safe depressurization and disposal of the used adsorber.**

Safe disposal of the used adsorber

A **venting adapter** is included with the new adsorber.

- **Attach** the **adapter** to one of the **self-sealing couplings** on the used adsorber.
- **Vent** the used adsorber to **atmospheric pressure.**
- Give the **used adsorber** to a **disposal company.**

2.8 Computer MRC (MRAWP) - MRSC (MRWP)

Required time: 10 min

Required tools: Vacuum cleaner

PM Cleaning the computer

MRC

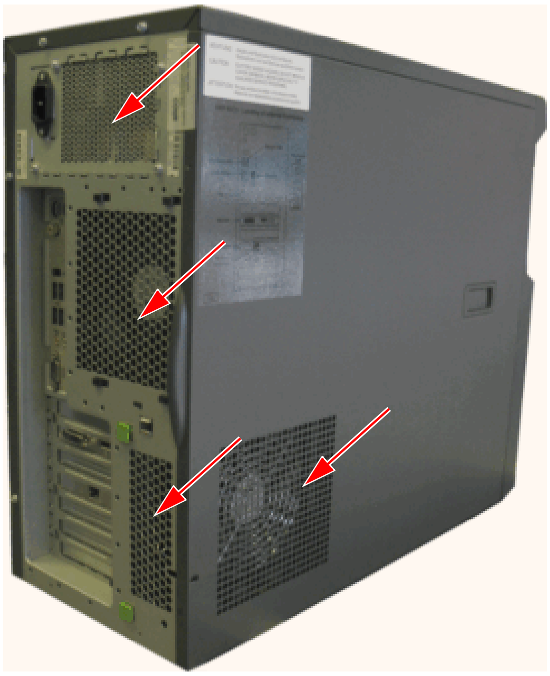
- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

MRSC if installed

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

Fig. 45: Computer

Fig. 46: Computer



2.9 TFT monitor

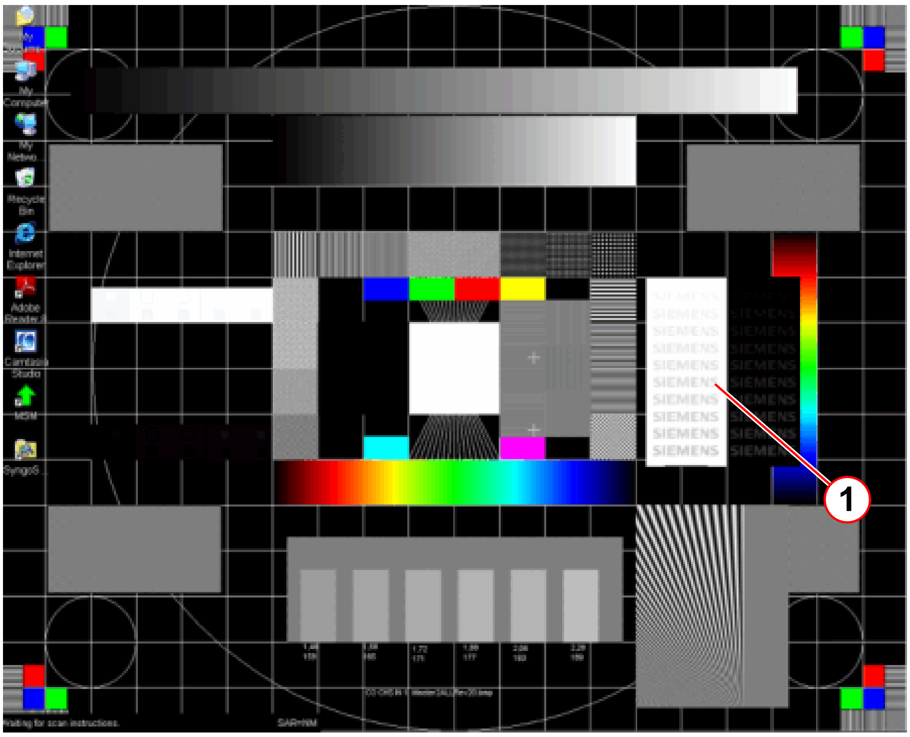
Required time: 10 min

Required tools: n.a.

2.9.1 Calling up the test image

- Simultaneously press the **Ctrl** and **Esc** keys.
  - » the Windows **Start Menu** appears
- Click: **Start > Programs > Advanced User** > enter PW > press OK button
- Simultaneously press the **Ctrl** and **Esc** keys.
- Click **Programs-> Accessories -> Windows Explorer -> Hard Disk C -> Med Com -> Utils -> Image quality**
- Double click the file name: Master2AllRev20.bmp
  - » test image opens
- Click full screen mode or press F11

Fig. 47: Test image - master2AllRev.20



(1) Siemens test logo

2.9.2 Check

PMA Checking the TFT monitor

- Check the MRAWP (MRC) TFT monitor using the test image.
  - » All colors are displayed
  - » Gray scale is correctly displayed
  - » Geometry is correct
  - » 10 SIEMENS test logos are visible in the black and white field
- If the MRWP (MRSC) TFT monitor (option) is installed, check the MRWP TFT monitor following the same procedure.
- If the In-Room MRWP TFT monitor (option) is installed, check the In-Room MRWP TFT monitor following the same procedure.

If any adjustment is necessary, refer to 'Replacement of Parts REP' Monitors; 'Visually inspecting/evaluating the TFT monitor' in detail.

2.9.3 Set the normal mode

- Press the **ESC** key to close the full screen mode.
- Close the program and the file explorer.

2.10 Patient Table

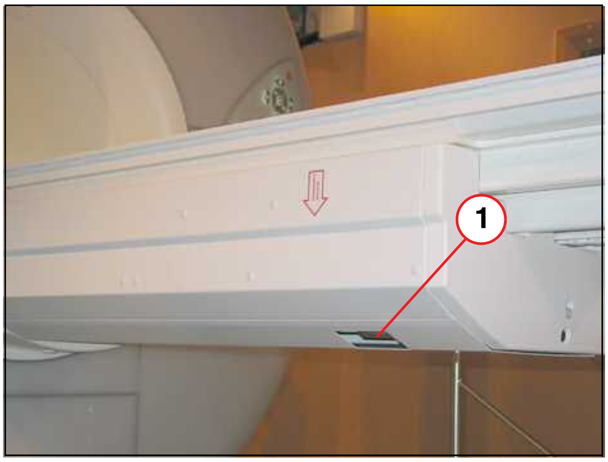
2.10.1 Patient Table - Drive Unit

Required time: 15 min

Required tools: 4mm Allen key

PM Checking the patient table drive unit

Fig. 48: Patient table



(1) Emergency release

There are different types of drives mechanisms for the patient tables (whole body and partial body). To differentiate the table type, check that there is an emergency unlocking device available (→ 1/Fig. 48 Page 42).

Whole body type:

» The emergency unlocking device is available.

Partial body type:

» The emergency unlocking device is not available.

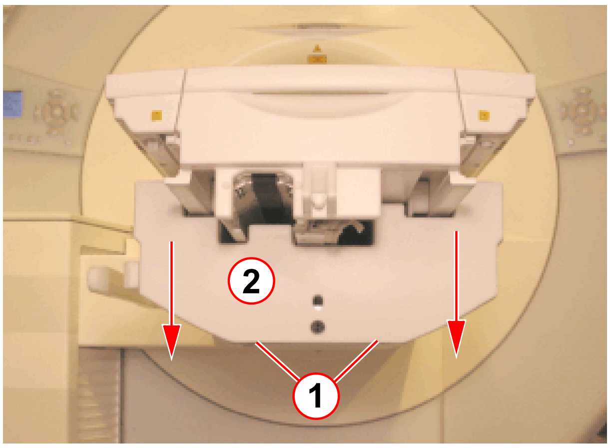
The toothed belt tension **must be** checked for the **partial body type only.**

For patient tables with **whole body drive unit** there is **no need to check** the toothed belt tension. Mark OK in the protocol.

2.10.1.1 Check the toothed belt tension for the partial body drive unit

- Move the table top in **Home Position**

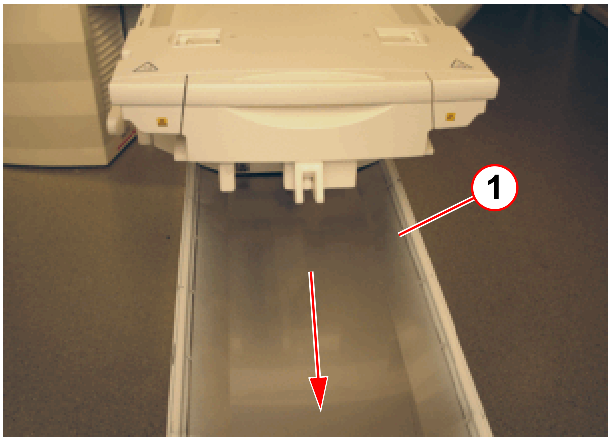
Fig. 49: Patient tabletop support frame foot end cover



(1) 4 mm allen screws
(2) Foot end cover

- Remove the 2 screws on the underside of the foot-end cover(1).
- Remove the foot-end cover (2) by sliding down.

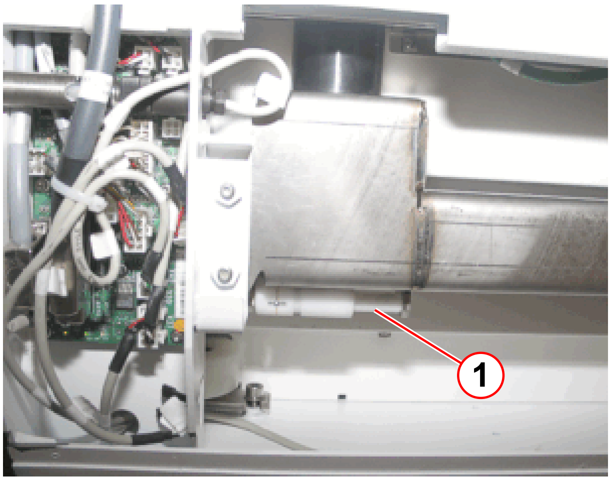
Fig. 50: Patient table support frame bottom cover



(1) Bottom cover

- Remove the bottom support frame cover (1) by sliding out.

Fig. 51: Support frame; bottom side

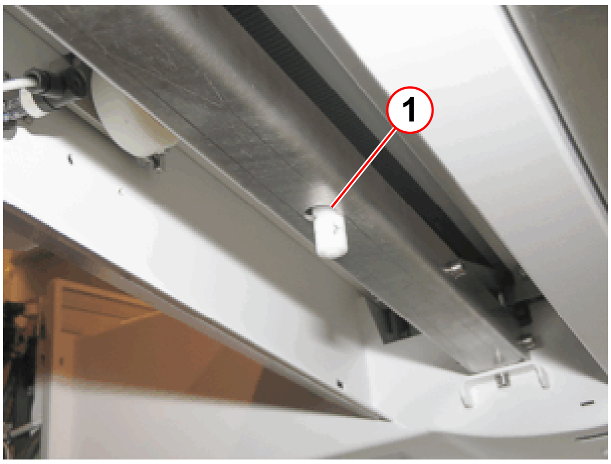


(1) Storage position of the service tool - toothed belt tension

The toothed belt tension service tool is located under the table top in the support frame.

- Remove the toothed belt tension service tool.

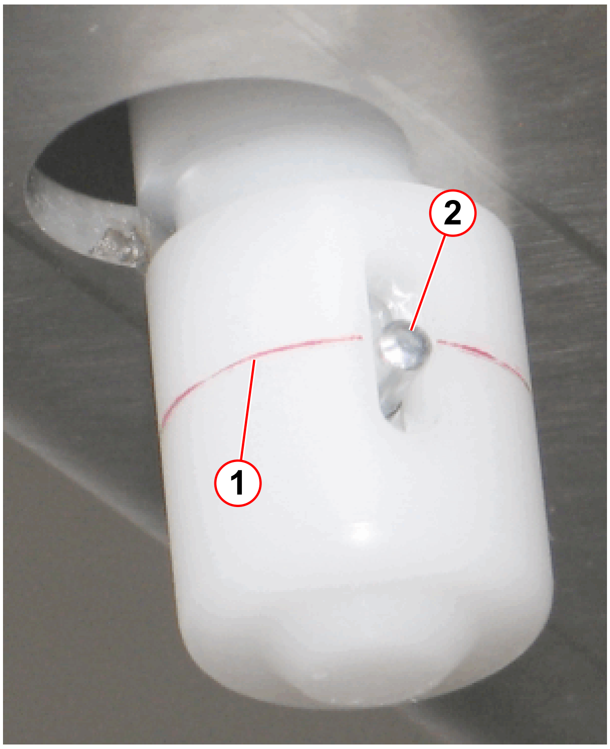
Fig. 52: Support frame bottom side



(1) Service tool - toothed belt tension in use

- Insert the toothed belt tension service tool (1) in the hole on the support frame bottom side, refer to (➔ Fig. 53 Page 44).

Fig. 53: Service tool - toothed belt tension



(1) Indication pin
(2) Ring marking

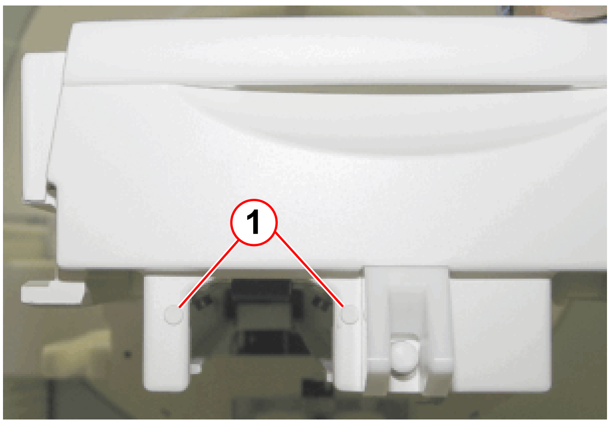
The indication pin shows the belt tension.

» The indication pin must be aligned with the marking ring.
» If the indication pin does not align with the ring marking, adjustment is necessary; refer to (➔ Adjustment of the toothed belt tension / Page 44).

- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to (➔ Fig. 51 Page 43).
- Reinstall the bottom support frame cover, refer to (➔ Fig. 50 Page 43).
- Reinstall the foot-end cover, refer to (➔ Fig. 49 Page 42).
- Mark OK in the protocol.

2.10.1.2 Adjustment of the toothed belt tension

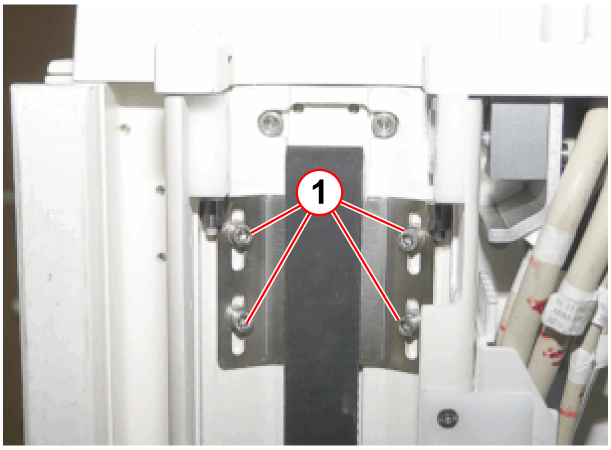
Fig. 54: Partial body patient table foot end side



(1) Blanking plug

- Remove the blanking plugs.

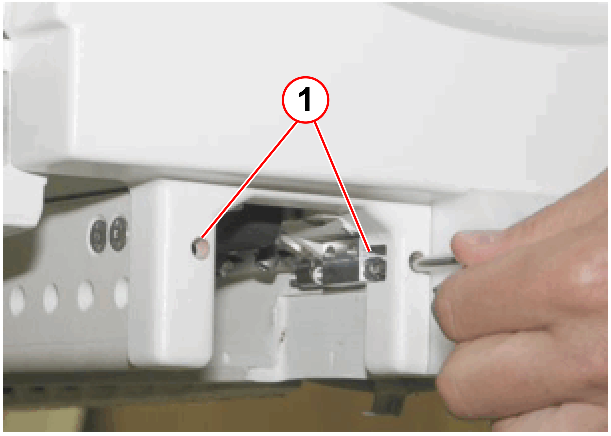
Fig. 55: Tension unit for the toothed belt



(1) Fixing screws (4 mm Allen screw)

- Loosen the fixing screws, do not remove.

Fig. 56: Toothed belt tension adjustment



(1) Adjustment screws for the belt tension

- Adjust the toothed belt tension with a 4 mm Allen key.
- Turn both adjustment screws (cw) equally until the indication pin is aligned with the marking ring, refer to (➔ Fig. 53 Page 44).
- Tighten the fixing screws, refer to (➔ Fig. 55 Page 45).

- Check the toothed belt tension, refer to (➔ Fig. 53 Page 44).
- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to (➔ Fig. 51 Page 43).
- Insert the blanking plugs, refer to (➔ Fig. 54 Page 44).
- Reinstall the bottom support frame cover, refer to (➔ Fig. 50 Page 43).
- Reinstall the foot-end cover, refer to (➔ Fig. 49 Page 42).

2.11 Patient Trolley

2.11.1 Patient Trolley (option)

Required time: 10 min

Required tools: 2 batteries (type “AA”)

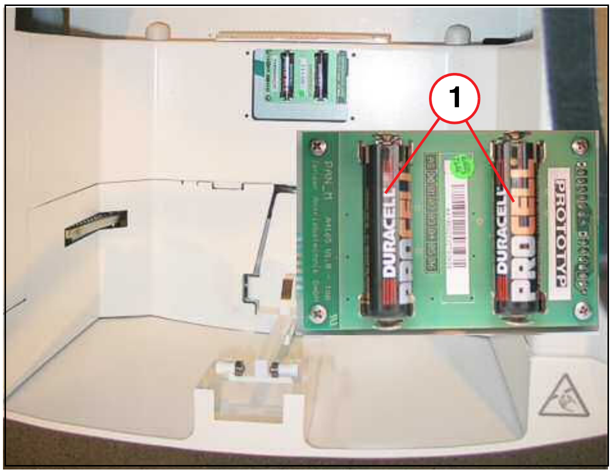
PM Replacing the patient trolley battery (option)

Fig. 57: Patient trolley

(1) Battery cover

Remove the battery cover.

Fig. 58: Patient trolley



(1) 2 mignon batteries

- Replace the batteries (battery type “AA”).
- Mount the battery cover.
- Turn the old batteries over to a disposal company in accordance with local regulations.

2.12 Mobile - Maintenance

2.12.1 RD298 Shock-log

2.12.1.1 Pre-requisites

Maintenance procedures for downloading the RD298 shock-log and replacing the Lithium battery.

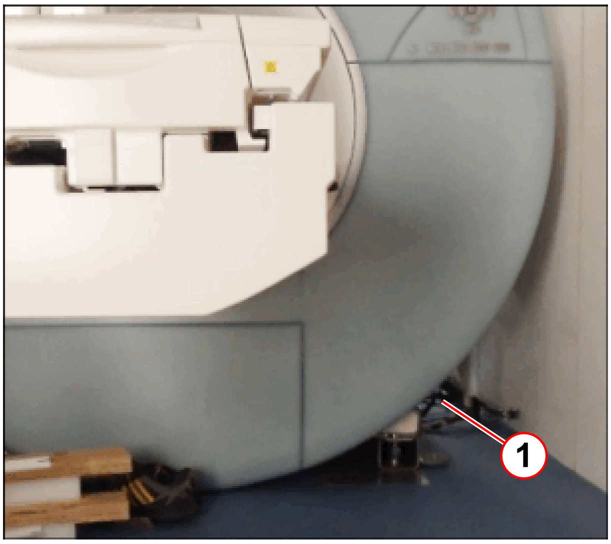
The battery must be replaced every 12 months.

Only use the lithium battery LTC size C, or LR14 (3.6V LS26500:SAFT). This battery is commercially available or can be ordered from CSML

If the Shock-log recorder contains data that you wish to preserve (and has a battery with some life in it), then download the data first. Any recorded data is maintained when the battery is removed.

Consult the help files embedded in the communications software for further support and troubleshooting.

Fig. 59: Shock-log location



(1) Location (covers removed)

- Remove the appropriate system covers to access to the RD298 shock-log unit.
- Put all fixing screws in a safe place when removed.

2.12.1.2 Time Required

Tab. 6 Time required

Process	Time required
shock-log Download	30 min
Battery Replacement	30 min

2.12.1.3 Tools Required

- Flat bladed screwdriver (Non-magnetic).
- Laptop PC and USB converter.
- Communications Software provided by Lamerholm Electronics.

2.12.1.4 Parts Required

- 2 x 3.6 V lithium battery AA type- (material number **10100429**).
- The battery is commercially available and can be sourced locally.
  » Recommended battery LTC size C, or LR14 (3.6V LS26500:SAFT). This battery is commercially available or can be ordered from CSML

2.12.1.5 Safety

CAUTION

Strong Magnetic Field

Damage to laptop PC and equipment.

» **When downloading data from the RD298 shock-log always use a suitable length USB cable. Keep Laptop PC away from the magnet.**

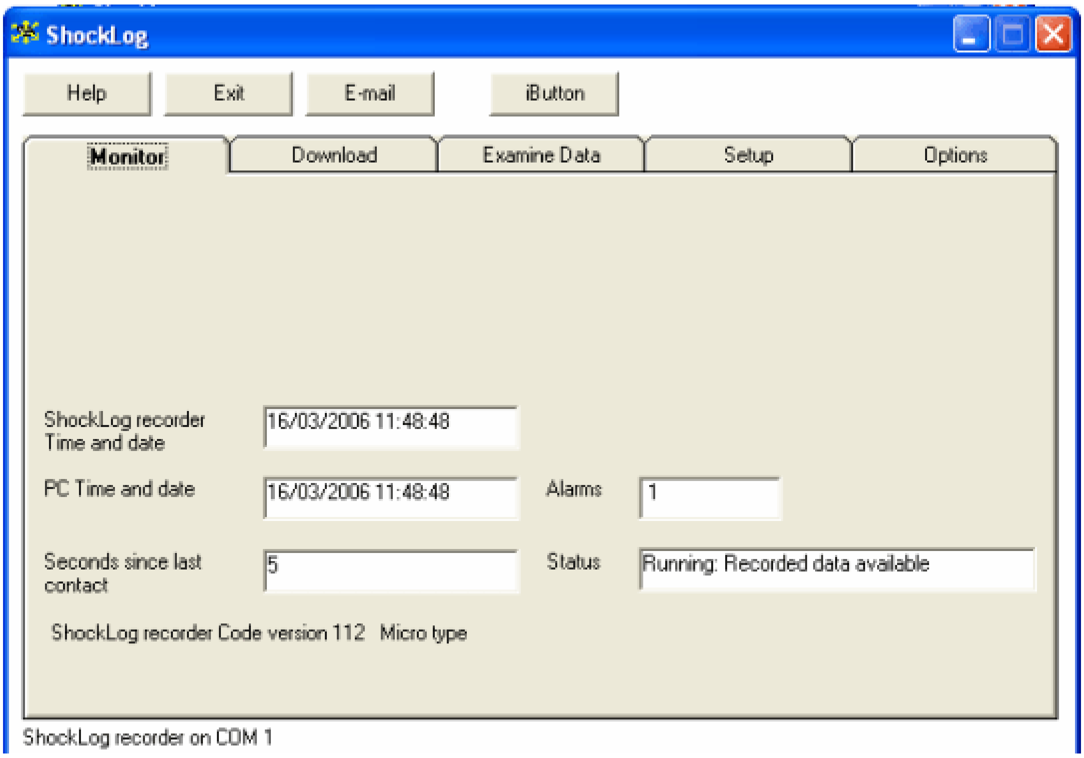
» **Use non-magnetic tools.**

2.12.1.6 Downloading the RD298 Shock-log Data

Connecting the Computer to the Shock-log RD298.

- In the equipment room, connect the 9-pin 'D' type or USB connector from the serial port on the Shock-log to the PC/USB port. (The connector will be labeled 'Shock-log').
- Run the Shock-log software, open the start-up screen and select Option 1.
  - The Shock-log working screen will load, with the Monitor Tab selected.
  - The bottom of the screen will display a status message 'Shock-log recorder on COM 1/2'/USB.

Fig. 60: Shock-log RD298 working screen



- The monitor display is updated once every 10 seconds, and the box labeled 'Seconds since last contact' reset to zero. **If this is not so, the unit is not communicating correctly**.

If this updating process is not operating, you must locate and correct the problem before attempting to perform any other communication dependent functions, i.e. check cable connections or battery condition.

Downloaded data can be examined even if the PC and Shock-log recorder are not communicating.

- Record the number of Wakeup/Alarms shown in the system log.

PM Downloading the Shock-log data

- Select the 'Download' tab and click on the 'Download data from Shock-log' button.
  - The dialogue box 'Enter user file name' will appear.
- Enter a file name, system number, date and the name of the CSE.
- Press the 'RETURN' key or click the 'OK' button.
  - The download checklist will appear and the PC will commence download.

The status box displays the download progress. The communications status may appear in the status line at the bottom of the Shock-log window.

When the download is complete, a 'Data Downloaded' message box will appear, reporting the file name and location of the data. The size of the file will vary between 30k and 500 k/bytes, depending on how long the Shock-log has been operating.

> Check data integrity using the 'examine data' tab, before clearing the memory.

- Clear the logger memory using the 'Clear all data' button.
- Ensure the logger is 'running' after download is complete.

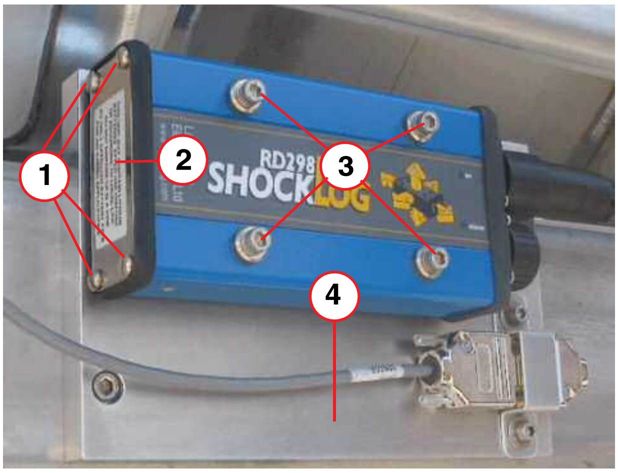
  This may take a few minutes. The logger may require a 'restart'.

2.12.1.7 Shock-log RD298 Battery Replacement

PM Replacing the Shock-log battery

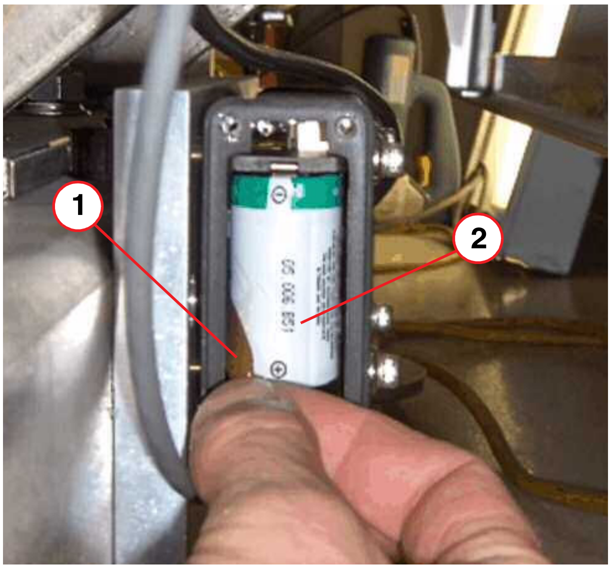
- Download and clear the data.

Fig. 61: Shocklog RD298 mounted on baseframe



(1) Battery cover screws
(2) Warning label
(3) Mounting holes
(4) Baseframe

Fig. 62: Shocklog RD298 Battery Removal



(1) Battery Tab
(2) Lithium Battery

> ⚠ **CAUTION**
>
> **Battery polarity must be correct.**
>
> **If incorrect, the battery will discharge very rapidly.**
>
> » **Check the polarity before fitting.**
>
> » **Do not use that battery for any application where a full battery life is required.**

- Remove the battery cover by undoing the 4 x screws at the base.
- Pull the plastic release tab to remove the battery.
- Check polarity and fit the new battery.

- Ensure the release tab is re-positioned correctly.
- Re-fit the battery cover using the 4 x screws.

Final Tasks

Re-setting the clock

- Reset the clock.
    - » At the start-up menu screen, select the download tab, followed by the ‘Set Clock’ button and follow the instructions.
- Record the replacement date of the new battery.
- Re-fit looks covers.

2.12.2 298™ Shock-log

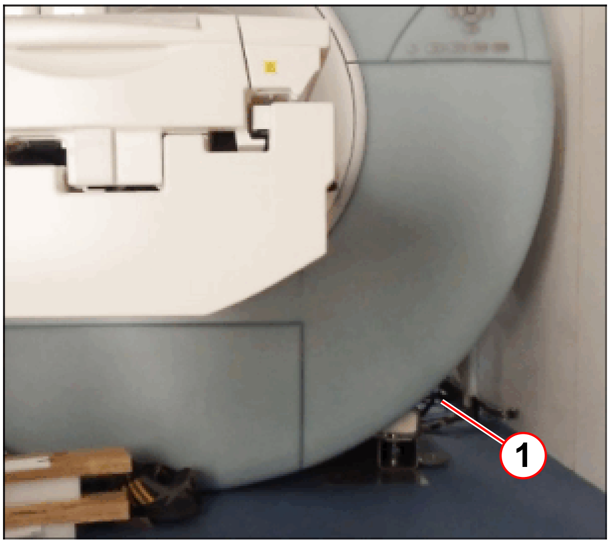
2.12.2.1 Pre-requisites

Maintenance procedures for downloading the shock-log™ and replacing the 2 Lithium batteries.

> The 2 batteries must be replaced every 12 months.
>
> Only use recommended 3.6V AA Lithium Thionyl Chloride batteries. This battery is commercially available or can be ordered from CSML

> If required download the shock-log™ data first.

Fig. 63: Shock-log location



(1) Location (covers removed)

- Remove the appropriate system covers to access to the shock-log™ unit.
- Put all fixing screws in a safe place when removed.

2.12.2.2 Time Required

Time required does not include setup and system cover removal.

Tab. 7 Time required

Process	Time required
shock-log Download	30 min
Battery Replacement	30 min

2.12.2.3 Tools Required

- Laptop PC and USB extender cable (41135M laptop end).
- Communications software for the 298 Shock-log™ provided by IMC Group Ltd
- Flat bladed screwdriver.
- Cross head screwdriver.
- 4 mm Allen (hex) key.

2.12.2.4 Parts Required

- 2 x 3.6 V lithium battery AA type- (material number **10622033**).
- The battery is commercially available and can be sourced locally.
  - » Recommended battery (Saft LS14500 Li-SOCl 3.6V AA).

2.12.2.5 Safety

⚠ CAUTION

Strong Magnetic Field

Damage to laptop PC and equipment.

- » **When downloading data from the shock-log™ always use a suitable length USB cable. Keep Laptop PC away from the magnet.**
- » **Use non-magnetic tools.**

2.12.2.6 Downloading the 298™ Shock-log Data

PM Downloading the shock-log™ Data

- Before changing the batteries download the stored data.
- Check that the latest software version is available.
- Connect the Shock-log™ to your computer with the USB cable. If necessary, use the USB extender cable.

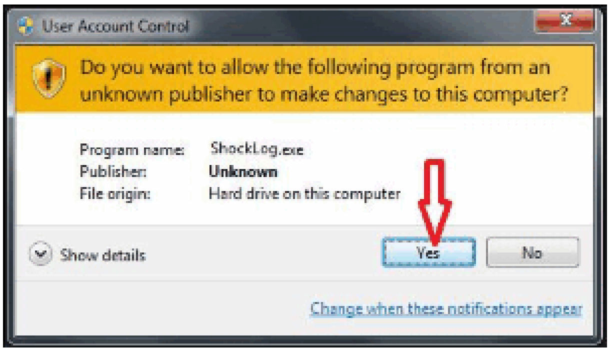


Fig. 64: Select Yes

- Right mouse click the icon and select **"Run as administrator"**.
- If the following window appears se-lect "Yes".

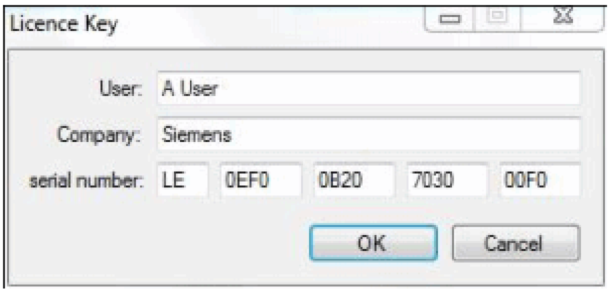
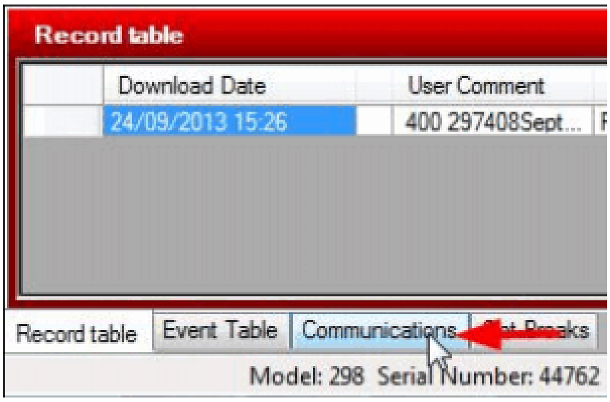


Fig. 65: Serial number

- Enter the following serial number when prompted:
  - » LE 0EF0 0B20 7030 00F0.

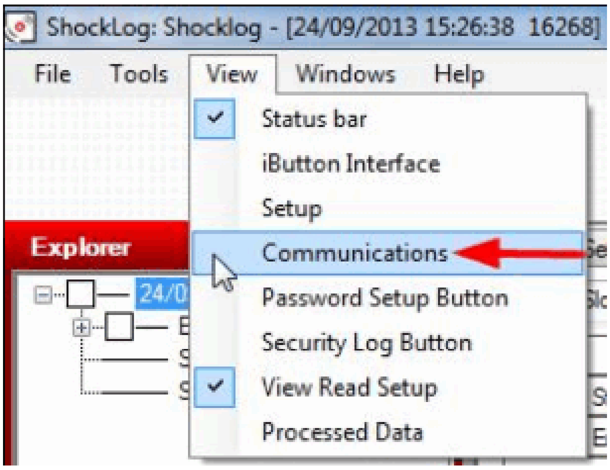
NOTE: 0 = zero

Fig. 66: Communication window



- Open “Communications” window

Fig. 67: Communications drop down

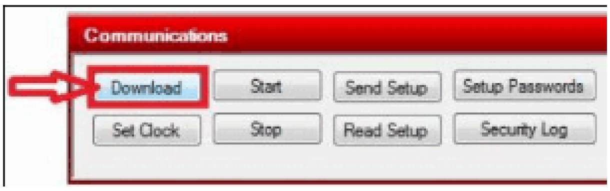


- The “Communications” window can also be accessed from the ‘View’ drop-down.

Fig. 68: Communications

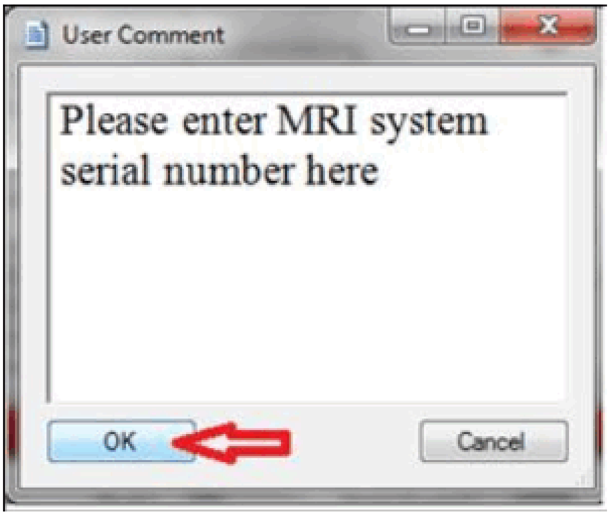
- In communication window select “Stop” button.

Fig. 69: Communication 'Download'



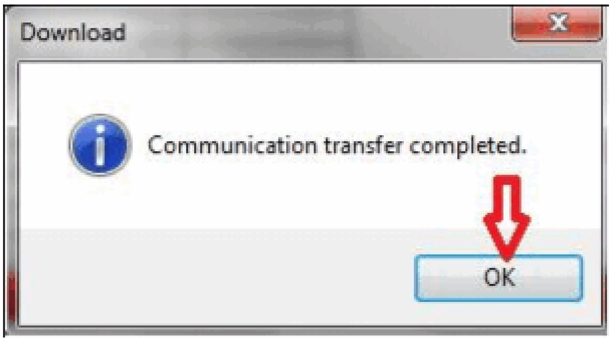
- After “State” changes to “Stopped” select ‘Download’ button.

Fig. 70: System serial number



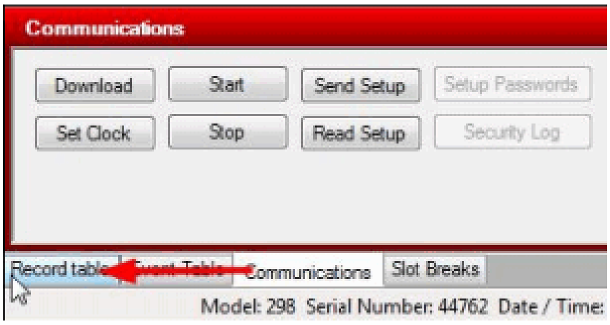
- Enter MRI system serial number in the following window, Press "OK".

Fig. 71: Transfer completed



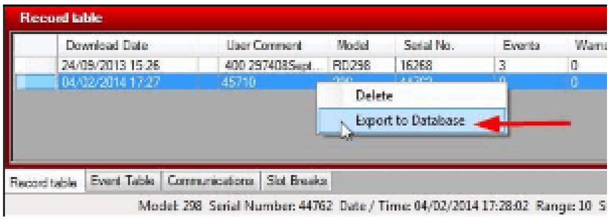
- After downloading the process is completed - Press "OK".

Fig. 72: Communications 'record table'



- Open "Record table" tab.

Fig. 73: Export to database



- Find downloaded file, right-click on it and select "Export to Database".

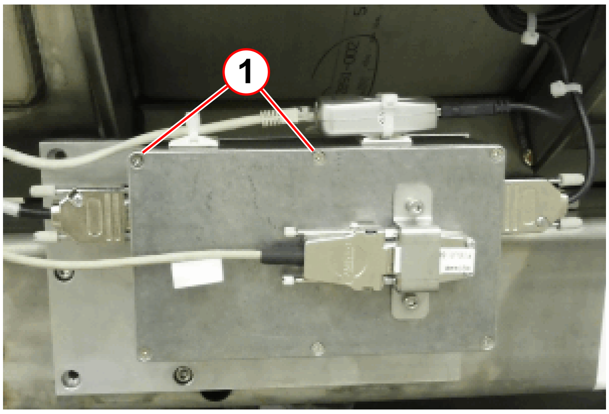
- Select 'Desktop' and then press 'Save' button.

- Make sure that you have a new “*.accdb” file saved on your desktop.
- Close Shock-log™ software and unplug Laptop.

2.12.2.7 Shock-log 298™ Battery Replacement

PM Replacing the Shock-log Battery

Fig. 74: Aluminium shielding box

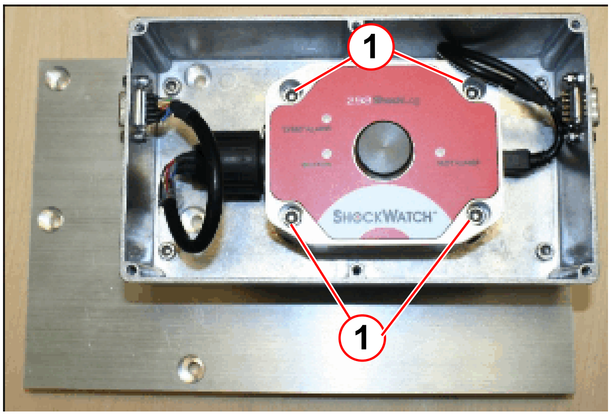


(1) Cover fixing screws x 6

It is not necessary to remove the cable connections to the aluminium shielding box.

- Remove the 6 x cross hd screws securing the shielding box cover.

Fig. 75: Fitting the Shock-log



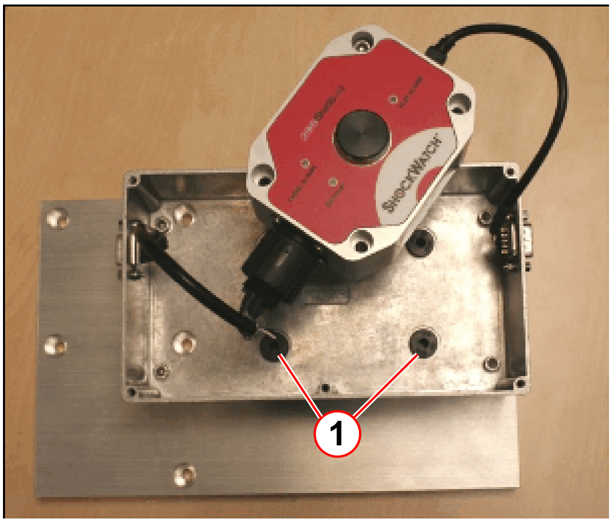
(1) 4 x M5 x 45 mm screws

To remove the shock-log™ from the aluminium plate:

- Remove the 4 x M5 x 45 St.St screws securing the shock-log™ to the mounting plate.

> It is not necessary to remove the inner RF box cables attached to the shock-log™ .

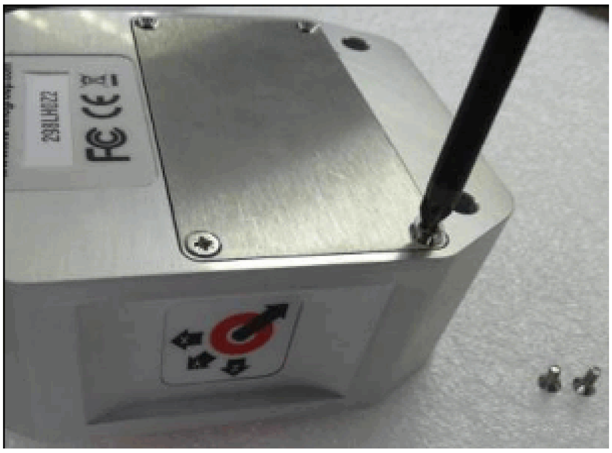
Fig. 76: Removing the Shock-log



(1) Rubber bush (x4)

- Carefully lift the shock-log™ from the RF box.
- Rotate the unit to access the battery cover on the back.
- Take care not to lose the 4 x rubber bushes from the aluminium back plate.

Fig. 77: Battery cover



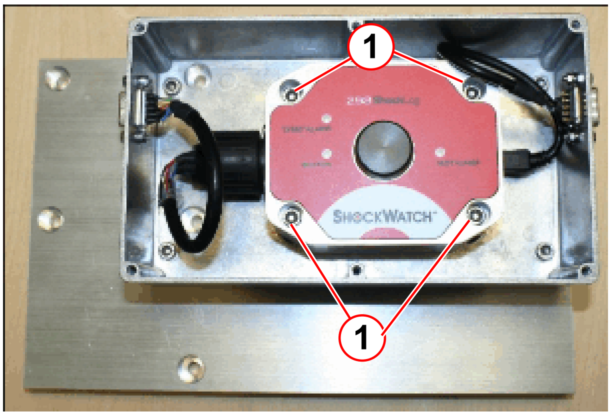
- The battery compartment is located under the plate on the bottom face of the shock-log™ .
- Remove 3 x screws.
- Undo the 4th screw but leave attached to the shock-log™.
- Move the cover to access the 2 x Lithium batteries.

The batteries discharge very quickly if fitted incorrectly. Check the polarity before fitting.

Fig. 78: Battery fitting

- Check the correct polarity before fitting.
- Fit the 2 new 3.6V Lithium Thionyl Chloride batteries.
- Replace the cover and tighten the 4 x screws.
- Check that the shock-log™ is running – “SLOT ALARM” LED and blinking every 10 sec.

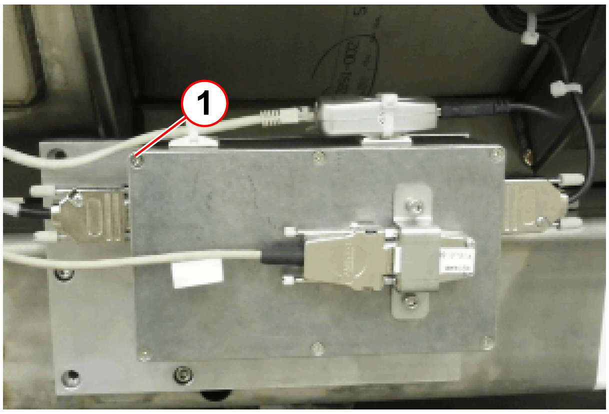
Fig. 79: Fitting the Shock-log



(1) 4 x M5 x 45 mm screws

- Fit the 4 x M5 x 45 St.St screws securing the shock-log™ to the mounting plate.
- Tighten screws, hand tight.
- Check that the shock-log™ is oriented (as shown) with the SL USB end oriented towards the magnet service end.

Fig. 80: Fitting the cover



(1) Fixing screws x 6

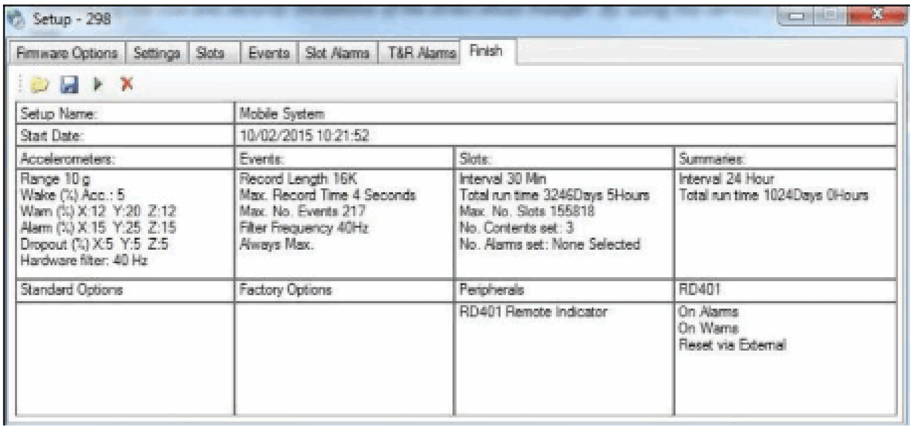
- Fit the shielding box cover with the 6 x cross hd screws.

2.12.2.8 Installing the 298 Shock-log™ Setup Software

> **Check that the "*.accdb" data files saved on your desktop.**
> **The following steps deletes all data from the Shock-log™!**

- The Shock-log™must be set up the same as the configurations shown.

Fig. 81: Shock-log settings



Tab. 8 Alarm and warning levels

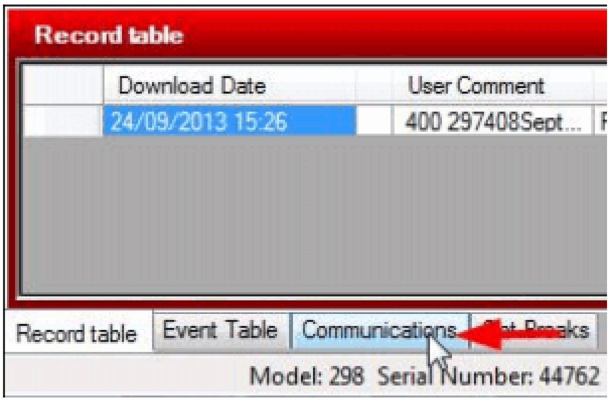
Acceleration Wake	0.5 g
Acceleration Alarm X (Axial)*	1.5 g
Y (Lateral)*	1.5 g
Z (Vertical)*	2.5 g
* Axis dependent on Shock-log™ orientation	

> The configurations in the mobile setup program are the definitive settings. The table above is only for reference.

The steps to install the set-up file onto a laptop are:

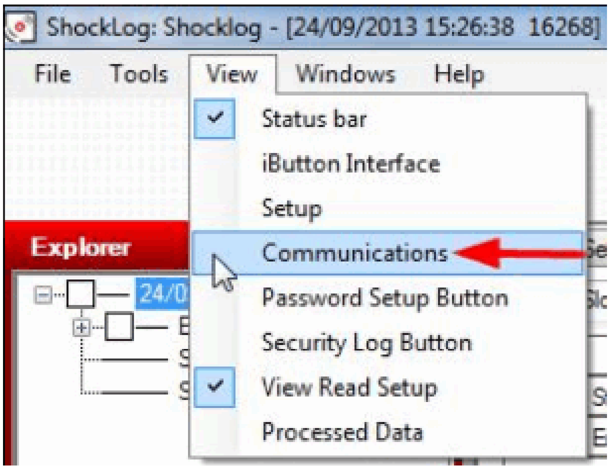
- Download the setup file "298 Mobile Setup V2.setup413" and save it on your desktop.
- Connect Shock-log™ to your laptop.
- Open Shock-log™ software.

Fig. 82: Open communications



- Open Communications window.

Fig. 83: Alternative drop down



- Alternatively open the communications window from the 'View' drop-down.

Fig. 84: Send 'setup'

- In Communications window select 'Send Setup' button.

- When the following window shows up, go to 'Desktop', and select the relevant setup file and select 'Open'.

Fig. 85: Select 'Desktop'

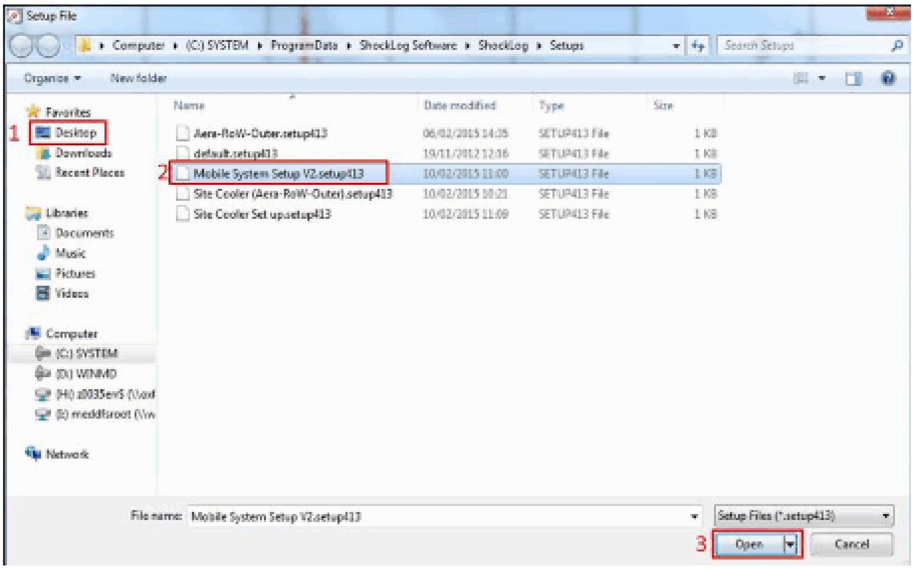
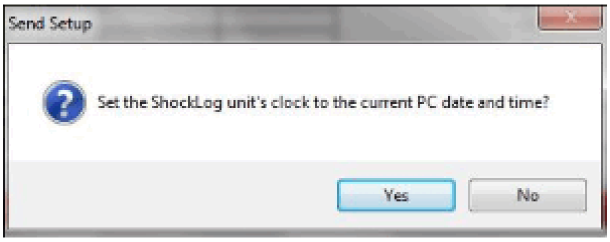
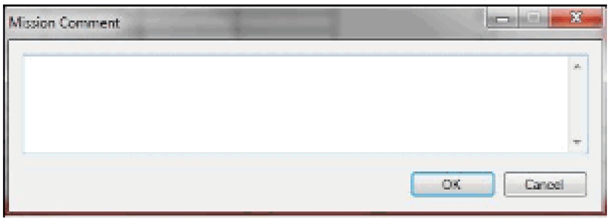


Fig. 86: Send clock setup



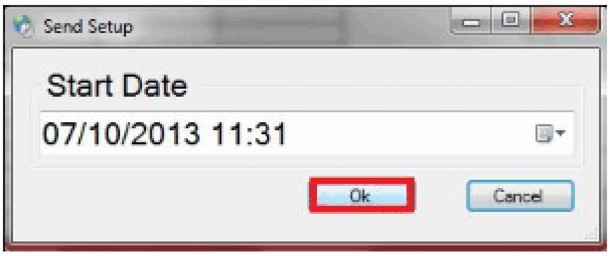
- When this window shows up select"Yes"

Fig. 87: Select 'OK'



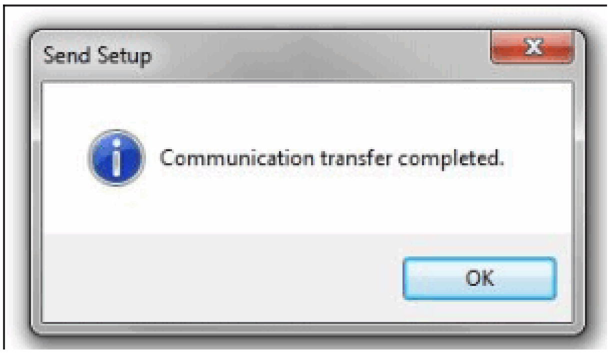
- Press 'OK'.

Fig. 88: Start date



- Press 'OK' again.

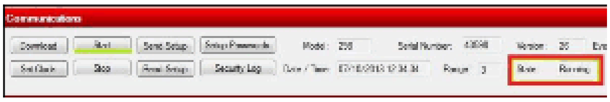
Fig. 89: Setup complete



- After setup is completed, the following window appears - select ‘OK’.

- After about a minute open communications window again and check that the state is “Running”.
- If it is in “Waiting” or “Stopped” state, press “Start” button.
- After 30 - 60-seconds the state changes to “Running”.

Fig. 90: Communications window



Final Tasks

- When you are sure that Shock-log™ is running, unplug it from the laptop and coil up the cable.
- Check that the 298 Shock-log™ power is switched on.
- Check that the permanent power supply unit to the 298 Shock-log™ is connected.
- Replace the system covers.
- Store the components with the magnet spare parts.
  - IMC Group Shock-log™ CD.
  - USB Extenders, laptop side (41135M).

2.13 Software

2.13.1 Cleaning up Directories

2.13.1.1 Cleaning up directories

Required time: 10 min

Required tools / parts: n.a.

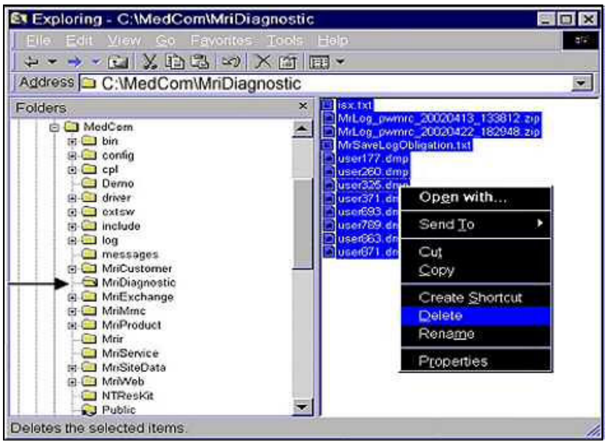
SW Deleting the MR save log files

Starting the Explorer

- Press <Ctrl-Esc>
- Select <**Start**>, <**Programs / All Programs**>, <**Advanced User**>, enter the Advanced password.
- Press <Ctrl-Esc>
- Select <**Programs / All Programs**>, <**Accessories**>, <**Windows Explorer**>

Delete files

Fig. 91: Exploring C:\MedCom\MriDiagnostic



- Navigate to" C:\MedCom\MriDiagnostic"
- Click once in the right window and select all files <Ctrl-a>
- Right-click one of the files and select <Delete>.
- This opens the pop-up window "Confirm Multiple File Delete"
- Click <Yes> to delete the files
- Close the Windows Explorer

> With software version VB19A, VB19A-SP3 and VB20P-SP4 embedded control, needs some adjustments.

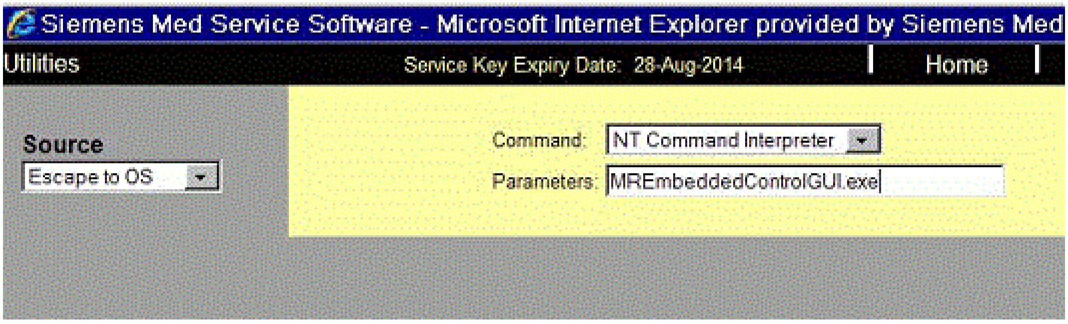
2.13.1.2 Switch embedded control to update mode

Action

- Select **Options** ⇒ **Service** ⇒ **Local Service**.
- Enter the valid service key and click **OK**.
- Start the **MREmbeddedControlGUI** tool from the local service UI under Utilities:
  - Under **Source** select `Escape to OS`
  - Under **Command** select `NT Command Interpreter`

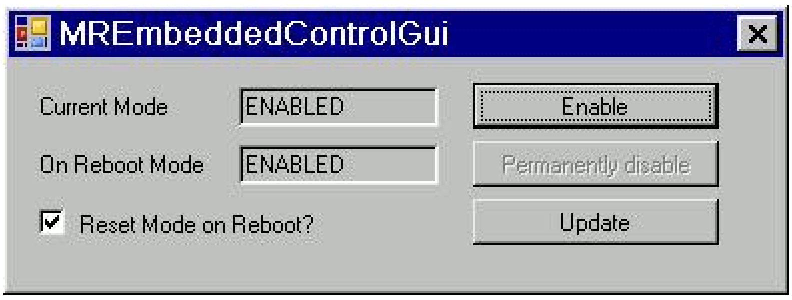
- Under **Parameters**, type in the following
  `MREmbeddedControlGUI.exe`
  and press **Go** to execute program.

Fig. 92: Execute MREmbeddedControlGUI



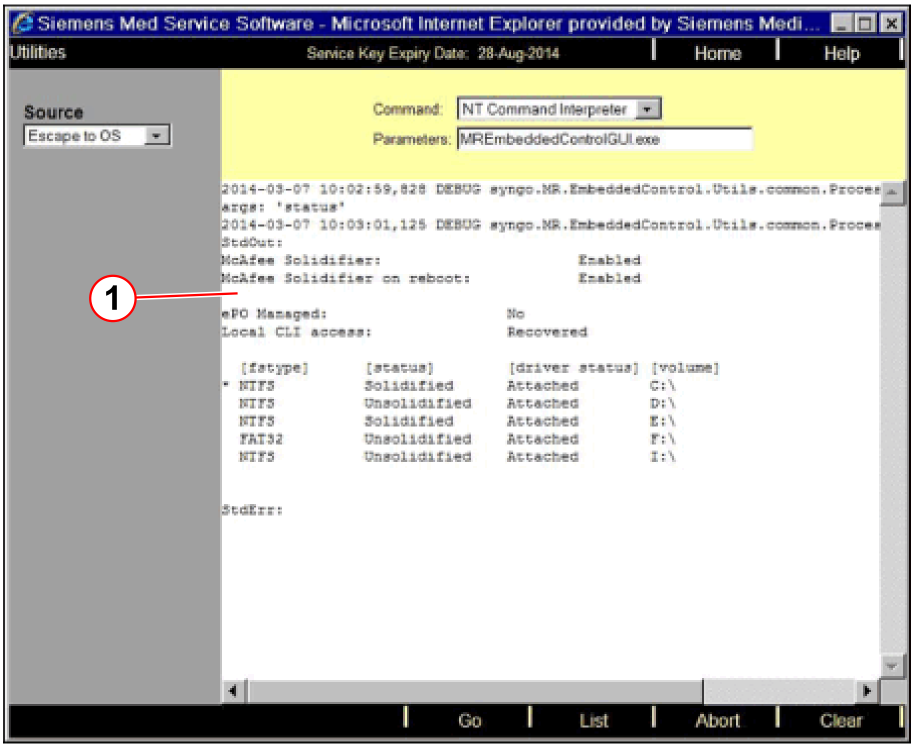
- The MREmbededdControlGUI is now displayed.

Fig. 93: MREmbeddedControlGUI



- The output log of the MREmbeddedControlGUI is shown in the Local Service UI (see figure below)

Fig. 94: Local Service UI



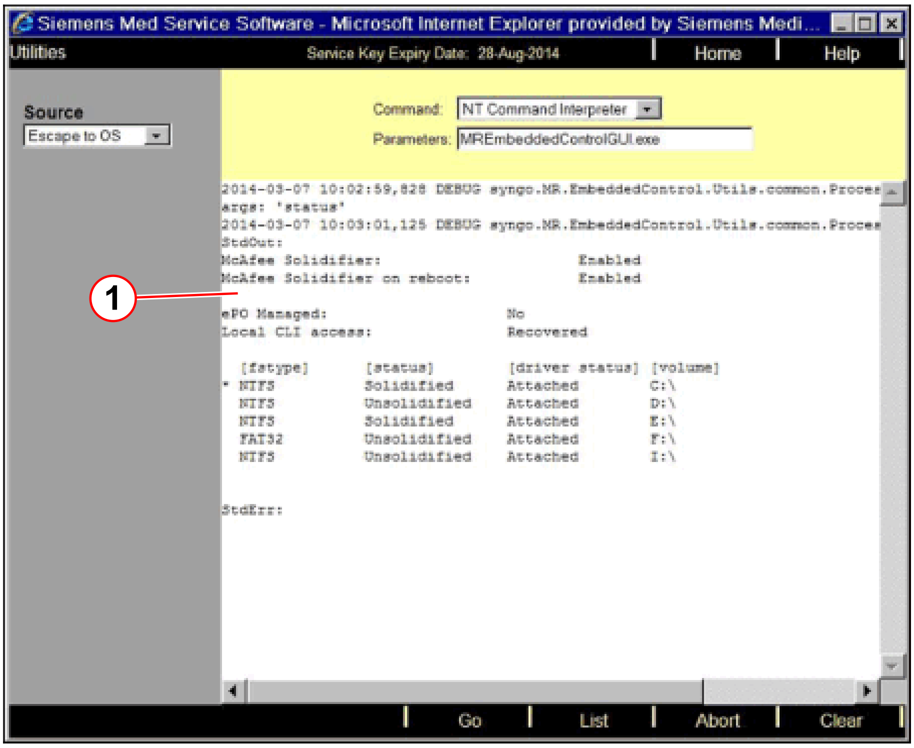
(1) Output log of the MREmbeddedControlGUI, the output above is only an example.

- Press the **Update** button to switch embedded control to update mode and confirm the displayed popup with **OK**.

  The update mode is now shown for the Current Mode.

- Press **Go** to start the MREmbeddedControlGUI again.

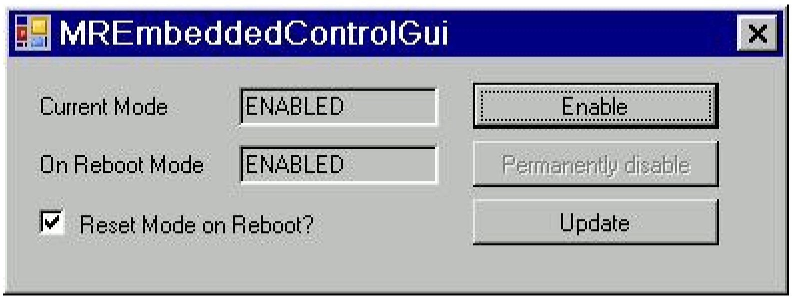
Fig. 95: Local Service UI



(1) Output log of the MREmbeddedControlGUI, the output above is only an example.

- The MREmbededdControlGUI is now displayed.

Fig. 96: MREmbeddedControlGUI



Press the **Enable** button and confirm the displayed popup with **OK**.

- To close the MREmbeddedControlGUI press the **X**.

2.13.2 Site-specific data

» Although site-specific data should be saved once a year, we recommend to perform it every three months.

SW Saving site-specific data

2.13.2.1 Software Version *syngo* MR B13 (Numaris/4 VB13A) or higher

Backing up site-specific Data on CD-R

Required time: 10-60 min

Required tools/parts: CD-R

> Use only medical grade CD-Rs, such as Siemens Medical CD-R 80 or TDK Medical CD-R.

Save the site-specific data each time you make changes to the site-specific configuration, tune-up or sequences and protocols. In the event of a hard disk crash, you will be able to restore a valid set of site-specific data.

- Login as **administrator**
- Start the service platform and select **Backup & Restore**.
- Insert a recordable CD into the CD/DVD-burner.
- Select **Backup** from the “Command” drop-down menu.
- In the Drives menu, select the target drive CD/DVD-burner (drive letter **S:**) to which you wish to save the data.
- Select **Master-Backup** from the “Packages” menu.
- Click **Go** to start the “Backup”. Check for errors in the log window after the backup has been completed.

Settings that are not part of the site-specific data:
Remote access mode (No Access, Limited Access, etc. in the customer UI).
Auto transfer flag (“Configuration”, “Service”, “Auto Transfers”)
Postscript Printer Configuration (customer paper printer)
ImageText Settings (image text settings from ImageText dialog)
Time zone
Transfer rules

> During backup at the MRSC, a “**CorRea.bru**” error may be displayed. Please ignore this error.

Backing up site-specific Data on external USB device

Required time: 10-60 min

Required tools/parts: USB device

> Please check the external medium (e.g. USB-disk) first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Log in as **administrator** ↵.
- Start the "Internet Explorer".
- Connect the USB device.
- Check what drive letter was set for the USB device with the Windows File Explorer.
- Log in to the service platform and select **Backup & Restore**.
- Select Backup in the Command menu, select the drive letter of the USB device, e.g.,**F** in the Drives menu.
- Select **Master-Backup** in the Package menu, start the backup by clicking **OK**.
- (Estimated time ~10-60 min) Check for errors using the log window.
- Click **Home** to leave Backup & Restore.

2.13.2.2 Software Version *syngo* MR 2004 / 2005 / 2006 ... (Numaris/4 VB12 ...)

Saving site-specific data

Required time: 20 min

Required tools / parts: CD-R

> Use only CD-R 's with medical grade, e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

When you are required to reinstall software (e.g. because of a defective system disk), the set of dynamic data will help you put the system back in operation as quickly as possible.

For this reason, we recommend to save the site-specific data each time you make changes in the site-specific configuration, tune-up, or sequences and protocols, even if the three-month interval has not expired.

Settings that are not part of the backup/restore

The following settings are not part of the site-specific data. Make sure you keep the settings in your configuration pages.

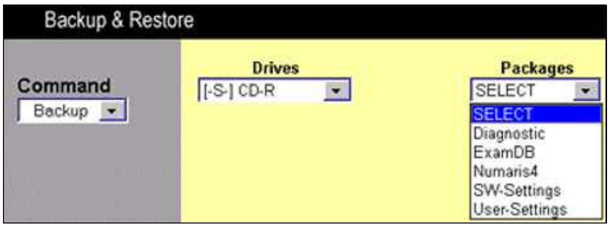
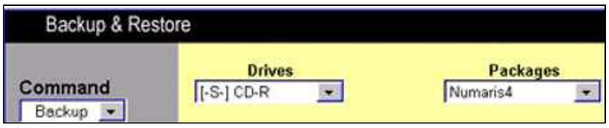
-
  - **DB offset**
  - **Printer installation, printer configuration**

> In case the error message "C:\MedCom\MriSitedata\ContextVision\.siemensIRU-PCNT-init-> access denied" is displayed during restore of data, re-enter the cv-license as described in 'Installation Software'.

- Start the service platform

- Select **Backup & Restore**

Tab. 9

Output	Input	Note
Fig. 97: 	Select **Backup** in the Command pull-down menu	Select Backup
Fig. 98: 	Select backup device, e.g. **CD-R** in the Drives pull-down menu	**CD-Burner** is associated with drive letter **S**
CustomerProtocols	Select **CustomerProtocols** in the Packages pull-down menu Click **Go**	Estimated backup time is ~1 minute
	Wait for message: "End Backup"	
SW-Settings 02 *Fig. 99:* 	Select **SW-Settings 02** in the Packages pull-down menu Click **Go**	Estimated backup time is ~4 minutes
	Wait for message: "End Backup"	
Numaris4 *Fig. 100:* 	Select **Numaris4** in the Packages pull-down menu Click **Go**	Estimated backup time is ~20 minutes
	Wait for message: "End Backup"	

Output	Input	Note
Security Settings **Only valid for systems with HIPAA**	Select **Security-Settings** in the Packages pull-down menu Click **Go**	Estimated backup time is ~1- 5 minutes
	Wait for message: "End Backup"	

2.14 End of Maintenance Instructions

The service covers which have been removed during safety checks have to be reinstalled before handing over to customer.

Initial Version

These instruction and the (→ Safety-Related Tests according to IEC 62353 / M6-020.833.05) replace the following document:

- Maintenance Instructions Combined Workflow
- The maintenance instruction (these document) (→ M6-020.831.05 / Preventive Maintenance / Page 1)
- Safety- related test (→ Safety-Related Tests according to IEC 62353 / M6-020.833.05)

Chapter	Changes
All	The procedures in these instructions are taken over from the document 'Maintenance Instructions Combined Workflow' with minor changes. ▪ TFT monitor: Switch embedded control to update mode, added. ▪ Editorial changes

Version 02

Chapter	Changes
Maintenance Instructions	Mobile maintenance: (→ Mobile - Maintenance / Page 47) ▪ Shock log procedure added.

Version 03

Chapter	Changes
Maintenance Instructions	▪ QA and RF cabin door maintenance deleted. Notes regarding QA and RF Cabin procedure added. ▪ TFT monitor test revised. (→ TFT monitor / Page 40) ▪ Minor editorial changes.

There are no Hazard IDs in this document.

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Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

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